

End-to-end MLOps framework for EEG-based epilepsy prediction: Automating deep learning pipelines for scalable healthcare solutions

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ABSTRACT

Predicting epileptic seizures from EEG signals is a critical yet challenging task, especially when transitioning machine learning (ML) models from laboratory settings to clinical environments. Most existing approaches lack adaptability to patient variability or remain confined to experimental contexts, relying on manual pipelines with limited automation. They generally do not support deployment and are seldom validated under real-world conditions, limiting scalability and clinical applicability. To address these limitations, we propose an end-to-end MLOps framework specifically designed for EEG-based seizure prediction. Our solution leverages the CRISP-ML(Q) methodology to manage the full ML lifecycle, including automated preprocessing, cross-validation, patient-specific model training, hyperparameter optimization, scalable deployment, and real-time monitoring. The predictive engine integrates a one-dimensional Convolutional Neural Network (CNN) with a Bidirectional Long Short-Term Memory (BiLSTM) network and an attention mechanism to capture spatial-temporal EEG dynamics. Validated on the CHB-MIT EEG dataset, the model achieved an average accuracy of 99.0% and sensitivity of 99.4% on training and test datasets, demonstrating strong generalization and adaptability to inter-subject variability. Following training, the MLOps pipeline autonomously selected optimal configuration parameters, including processing and model architecture, yielding performance improvements exceeding 15% compared to static baselines. In production, the deployed system remained fully operational under intensive concurrent usage, with consistent inference latency below one second in remote conditions. A user-friendly, real-time dashboard enables continuous monitoring, seizure alarm triggering, and live visualization of model behavior and evaluation metrics. These results confirm the robustness, scalability, and clinical readiness of our MLOps-driven framework for real-time EEG-based seizure prediction.

1. Introduction

In neurology, two main techniques are used to assess brain function and structure: anatomical and functional. Anatomical techniques provide images of brain structure, while functional techniques provide insight into changes in brain activity over time. Electroencephalography (EEG) is the most commonly used functional method, recording electrical activity from electrodes placed on the scalp (sEEG) or directly on the surface of the brain (iEEG). sEEG is more commonly used by neurologists because it can identify abnormal patterns, such as those found in sleep analysis, brain-computer interfaces (BCIs), and epilepsy diagnosis [1].

Epilepsy is the second most common neurological disorder, affecting about 50 million people worldwide. Early prediction and detection of

seizures can reduce the risk and significantly improve patient quality of life [2,3]. Thanks to the work of Gibbs and Gibbs in 1936 [4] different wave frequencies and patterns can now be associated with brain states. The four basic phases of the EEG are interictal (between seizures), preictal (before the seizure), ictal (during the seizure), and postictal (after the seizure). In the preictal phase, neurons often synchronize abnormally, signaling an impending seizure. Historically, EEG diagnostics have relied on visual observation, which is error-prone and time-consuming. As a result, reducing the possibility of early intervention. IoT and ML now play a central role in EEG epilepsy diagnosis, automating processing and improving classification accuracy [5–7].

Despite these advances, many approaches remain too complex for practical implementation. Processes require human intervention and

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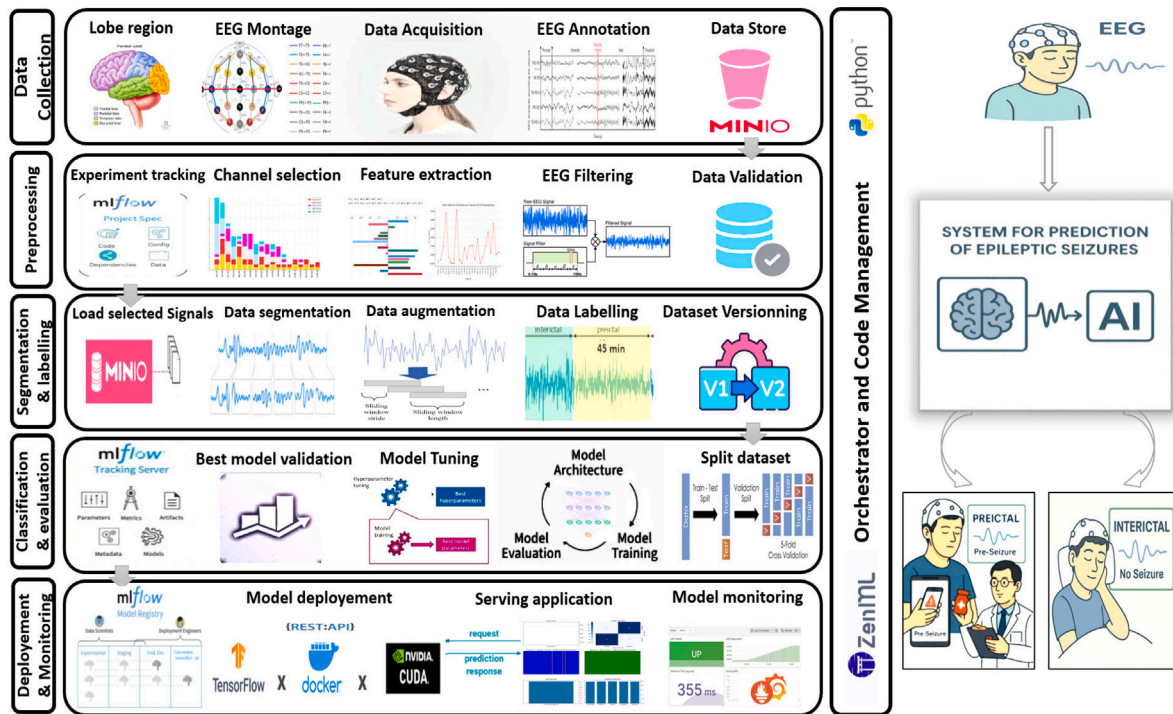


Fig. 1. Overview of the proposed MLOps-driven pipeline for EEG-based seizure prediction system that differentiates between pre-seizure and normal states to guide clinical decision-making.

delay decision-making. To overcome these challenges, organizations have adopted MLOps workflows to support scalable deployment of ML models. MLOps standardizes the model lifecycle from idea to inference. automates the entire lifecycle, including data preparation, training, evaluation, and real-time monitoring, ensuring adaptability over time. It also monitors patient data variability and reduces the risk of model drift. MLOps enhances robustness, minimizes errors, and enables accurate, real-time predictions. Furthermore, MLOps tools contribute to explainability, helping clinicians interpret and trust outputs [8,9]. Hence, by introducing operational discipline into machine learning pipelines, it ensures that experimental models can evolve into maintainable clinical solutions [10].

The Array journal covers various machine learning applications in healthcare, such as seizure detection using dual-stream Transformers that combine time and frequency features [11], and seizure prediction using LSTM-based recurrent models [12]. It also includes works on attention CNN-LSTM model for COVID-19 detection from chest X-rays [13]. These studies show the journal's focus on data-driven medical diagnostics and efficient model design. However, most of the work stays at the experimental level, with limited attention to automation, deployment, traceability, or explainability.

In this work, we present an operational end-to-end MLOps framework for patient-specific EEG-based epilepsy prediction. It bridges gaps between research prototypes and real-world services. Our AI-driven workflow is fully automated from data preprocessing to model deployment and real-time monitoring. Our approach follows the CRISP-ML(Q) methodology [14], offering structured, quality-driven pipelines. At its core, we designed a one-dimensional CNN-BiLSTM architecture with attention mechanisms, adapted to patient EEG characteristics. To enhance data quality, the automated preprocessing steps include filtering noise, removing artifacts, extracting features, and ensuring class balance. Personalization is further achieved through automatic channel selection and hyperparameter tuning. These models are deployed and integrated into real-time clinical dashboards to support epilepsy care. Such integration demonstrates the capacity of MLOps to unify experimental design, deployment logic, and clinical usability in a single

scalable environment. Each phase of the ML workflow is tracked and monitored, enabling clear insights into performance and explainability while ensuring reproducibility.

Fig. 1 provides an overview of the proposed framework, illustrating the pipeline components as well as a representative application scenario. In a real-time clinical setting, EEG signals are continuously processed by the deployed model to distinguish between interictal (normal) and preictal (pre-seizure) states. This predictive capacity, up to several minutes in advance, enables early intervention and supports timely clinical decision-making, thereby enhancing patient care. The integration of prediction, interpretability, and operational deployment demonstrates the potential of MLOps to unify experimental design, deployment logic, and clinical usability in a scalable healthcare-oriented environment.

The remainder of the paper is organized as follows: Section 2 reviews related work, while Section 3 introduces the shift from manual processes to an automated MLOps-driven EEG workflow. Section 4 describes the CRISP-ML(Q) methodology adopted in our approach, followed by Section 5, which details the implementation of the proposed MLOps framework. Section 6 presents and discusses the experimental results. Finally, Section 7 concludes the paper with key insights and perspectives.

2. Related work

This section reviews MLOps in healthcare, focusing on methodologies used to streamline the machine learning lifecycle. Although no existing research combines MLOps with EEG-based epilepsy diagnosis, several studies highlight its growing relevance in healthcare. The MLOps market, valued at \$839 million in 2023, is projected to reach \$17.97 billion by 2032, driven by the increasing adoption of ML technologies across industries such as healthcare, finance, and manufacturing [15]. Despite significant advances in AI-driven diagnostics, practical challenges persist namely in system integration, real-time monitoring, and scalable deployment. As emphasized by Ramírez et al. [16], MLOps is essential for bridging the gap between research prototypes and fully

operational systems. However, its integration into EEG-based seizure prediction workflows remains largely unexplored. Table 1 summarizes representative MLOps studies in healthcare, highlighting their key contributions and limitations.

2.1. Deployment-oriented frameworks

Deploying ML models in clinical settings requires scalable, private, and continuously integrated frameworks. MLOps helps bridge the gap between research and real-world use. Kannan et al. [17] proposed a federated, serverless MLOps architecture that enhances privacy and scalability but lacks a clear, reproducible process. Poddar et al. [18] introduced a stakeholder-focused framework for clinical ML adoption, yet did not include robustness or drift testing. Keshri et al. [19] developed a CNN-based PII detection system with Azure CI/CD, but its weak link to business goals limits broader use. Finally, Ng et al. [20] emphasized fairness through equity-driven MLOps principles, though their work remains mostly conceptual with few implementation details.

2.2. Trust, governance, and auditing

Building trust in clinical AI requires safety, transparency, and strong governance. MLOps can support this by embedding auditing and traceability into workflows. Helmer et al. [21] introduced an AI audit catalog into the MLOps pipeline, reducing 242 trust requirements to 82 tasks based on project maturity. Yet, their framework remains qualitative with no empirical validation. Rajagopal et al. [22] reviewed 148 studies on ethics, monitoring, and clinical integration but did not propose practical pipelines. Venkata et al. [23] showed how MLOps can improve patient care through real-time adaptive systems, though their work lacks detailed metrics and long-term scalability evaluation.

2.3. Monitoring, resilience, and drift handling

Healthcare ML systems must handle changing data and remain reliable under diverse conditions, making monitoring and resilience vital in MLOps. Gulyas et al. [24] used MLFlow to track scoliosis models in school screenings, but their setup lacks drift detection and fails in low-light settings. Viacheslav et al. [25] introduced resilience and uncertainty calibration to guard against adversarial threats, though they omit continuous data updates and real-world monitoring. Tembhekar et al. [26] reviewed drift detection techniques across healthcare and finance, yet their work remains theoretical with no empirical or deployment validation.

2.4. Applications to EEG and biomedical systems

Biomedical AI applications, such as EEG-based diagnosis, require not only accurate models but also structured MLOps pipelines for long-term use. El Menshawy et al. [27] created a real-time mobile system for epilepsy detection, but it lacks key MLOps features like automation, drift detection, and retraining. Gerogiannis et al. [28] proposed RACNet for COVID-19 diagnosis, focusing on scalability and ethics, though it was not tested in real clinical environments. Mallardi et al. [29] presented a full MLOps framework for healthcare, but its validation is limited to a single context without broad empirical testing.

In summary, MLOps are increasingly recognized as critical for operationalizing AI in healthcare. However, most studies lack complete coverage of the ML lifecycle, particularly in deployment monitoring, robustness, governance, and retraining. Notably, no existing research addresses EEG-based seizure prediction through a full MLOps pipeline. This highlights the importance and novelty of our work, which bridges this gap using a CRISP-ML(Q)-driven framework [14].

3. Diagnosis of epileptic seizures: From traditional process to MLOps workflow

3.1. Traditional process

Seizure classification involves data acquisition, preprocessing, and feature extraction. ML or DL techniques are trained to identify brain states, with results refined through post-processing for real-time detection and prediction. Traditional data acquisition systems are accurate but complex. Portable and wearable devices, such as Emotiv, Muse [7,30], and dry electrode systems like OpenBCI, simplify long-term recording. EEG-fNIRS hybrid systems improve detection accuracy by combining electrical and hemodynamic data [31]. Publicly available datasets, including CHB-MIT, Bonn University, and DREAMER, are invaluable for model training and testing [32].

Preprocessing techniques address noise and artifacts. Band-pass and notch filters remove unwanted frequencies, while ICA, EMD, and wavelet denoising manage artifacts [33,34]. Normalization, segmentation, PCA, and DWT reduce complexity and enhance feature extraction [31]. SMOTE is used to balance data between EEG phases [35]. Features are grouped into time-domain (e.g., mean, variance, skewness), frequency-domain (e.g., FFT, spectral entropy, power spectral density), time-frequency (e.g., STFT, Wavelet, Stockwell Transform), and nonlinear categories (e.g., entropy, fractal dimensions) [31]. Hybrid methods combining these features enhance ML models, while DL models like CNNs automatically learn complex patterns [6,33].

ML techniques such as SVM, KNN, Random Forest, Gradient Boosting, and XGBoost are widely used for EEG classification, leveraging manual feature extraction [36–39]. DL models, including CNNs, RNNs, LSTMs, GANs, autoencoders, and hybrid architectures (e.g., CNN-RNN), excel in automatic feature extraction, sequence learning, and dimensionality reduction [6,40–42]. Real-time operation strategies include cloud-based scalable resources [7], IoT integration with 5G NB-IoT [43], edge computing for local EEG processing [44], and hybrid cloud-edge systems for speed and efficiency [45]. Mobile and web applications provide accessible tools for clinicians and patients [27], ensuring flexible seizure prediction in real-world settings.

3.2. Automated MLOps workflow

ML pipelines consist of serialized steps that automate workflows, ensuring:

- Reproducibility: produces identical outputs for identical inputs.
- Automation: requires minimal manual intervention.
- Acceleration: enables faster transition from experimentation to production.
- Modality: supports the development of new solutions and components.
- Scalability: allows reuse of steps to integrate with external systems.

Most epilepsy diagnosis pipelines are at Google's "Level 0" of MLOps maturity [46]. This means they rely on manual processes and lack monitoring or feedback loops. These limitations can lead to issues like data drift and biased results. Google's MLOps maturity model has different levels of automation. Level 1 automates data processing, development, and evaluation. However, it does not include automated deployment or monitoring. Level 2 adds continuous feedback loops for automated updates and retraining. This reduces model drift and ensures scalable, real-time performance. Automated MLOps integrates data management, pipelines, models, hyperparameter tuning, experiment tracking, evaluation, integration, and deployment. It accelerates model production, improving speed, accuracy, and reliability. In practice, such automation transforms experimental ML workflows into resilient production systems capable of handling dynamic clinical data.

Table 1
Summary of representative MLOps studies in Healthcare.

Work	Domain	Key contributions	Limitations
[17]	Clinical AI deployment	Federated, scalable architecture	Lacks process model and QA
[18]	ML to clinical transition	Stakeholder-based deployment framework	No robustness or drift testing
[19]	PII detection	Azure-based CI/CD with transfer learning	Poor business-to-ML alignment
[20]	AI fairness in healthcare	Equity-centered MLOps principles	Largely conceptual, lacks implementation details
[21]	Governance	Maturity-based audit catalog for AI trust	No empirical metrics for validation
[22]	MLOps literature review	Ethical and regulatory landscape in healthcare AI	Lacks unified implementation pipeline
[23]	Real-time hospital analytics	Adaptive MLOps pipeline for patient care	Missing performance metrics and scalability validation
[24]	Scoliosis detection	MLflow tracking, real-world application	No drift adaptation, poor lighting performance
[25]	Imaging and safety	Resilience optimization, adversarial robustness	No lifecycle monitoring or data engineering
[26]	Healthcare & finance	Drift detection techniques review across domains	Theoretical only, no empirical deployment or validation
[27]	EEG-based epilepsy detection	Real-time monitoring and clinician support	No MLOps integration or deployment strategy
[28]	COVID-19 diagnosis	Ethical, scalable, explainable AI system	Not validated under real-world conditions
[29]	General healthcare MLOps	End-to-end MLOps deployment framework	Limited validation in narrow context

MLOps also simplifies model governance by ensuring traceability across data versions, model iterations, and deployment stages. By decoupling infrastructure concerns from experimentation, MLOps empowers researchers to focus on model improvement while ensuring operational stability at scale.

3.3. MLOps tools and platforms

End-to-end MLOps platforms offer a wide range of tools and capabilities. It aligns with software engineering best practices for ML systems. The key components include the following [47,48]:

1. **Data Management & Preprocessing:** Tools such as Great Expectations, Talend, and MinIO support data ingestion, storage, preprocessing, labeling, versioning, and integration with storage backends.
2. **Experimentation & Model Development:** Platforms like Mlflow, neptune.ai, and Comet ML facilitate model training, feature engineering, fine-tuning, and comprehensive performance evaluation.
3. **Model Deployment & Serving:** Solutions like BentoML, NVIDIA Triton, and Seldon Core enable efficient deployment via containerization and scalable API management.
4. **Model Monitoring & Performance Tracking:** Tools such as Evidently AI, Arize AI, and WhyLabs provide real-time model monitoring, metric tracking, anomaly detection, and stability evaluation.
5. **Collaboration & Version Control:** Systems like DVC and LakeFS ensure collaborative workflows, reproducibility, and artifact versioning.
6. **Automated Pipelining & Workflow Orchestration:** Workflow tools like ZenML, Kedro, and Prefect manage task scheduling, dependencies, and error handling, enabling complex ML pipeline automation.
7. **Model Governance & Compliance:** Arthur AI and Fiddler AI address privacy, fairness, explainability, and regulatory compliance, supporting responsible AI development.
8. **Integration with ML Tools & Libraries:** Popular frameworks like TensorFlow, PyTorch, and Scikit-learn are supported to enhance productivity and scalability.

These platforms offer advanced features such as parameter tuning, feature stores, GPU-accelerated cloud servers, and vector database tools. For large language models, specialized tools streamline training, fine-tuning, and validation. Such tools enable MLOps platforms to accelerate experimentation, enhance scalability, and ensure robust model monitoring. They are essential components of modern machine learning workflows. Such tools speed up ML experimentation, enhance scalability, and ensure solid model monitoring. MLOps platforms unify the ML lifecycle, reducing complexity and improving reproducibility and continuous delivery. They let research teams focus on experimentation while handling infrastructure, compliance, and deployment. Essential for modern ML workflows.

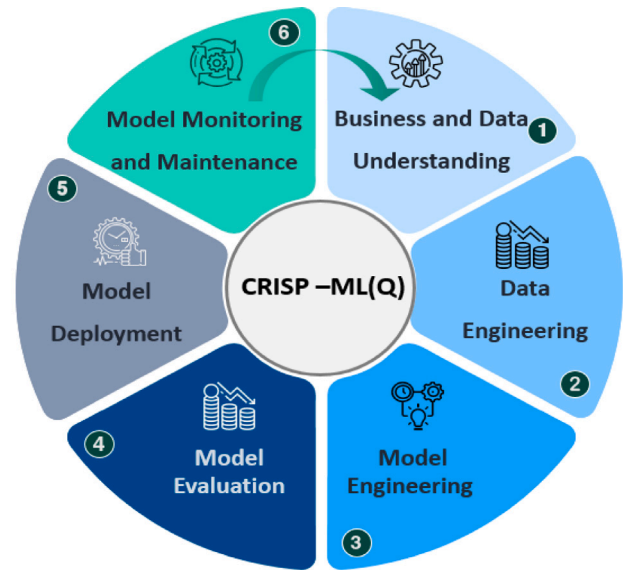


Fig. 2. Overview of the CRISP-ML(Q) process, illustrating the six key phases of a machine learning project. The cycle emphasizes iteration and quality throughout the ML lifecycle.

4. CRISP-ML(Q) methodology

MLOps boosts AI deployment agility, especially in critical settings. It is more effective with CRISP-ML(Q), a structured methodology. CRISP-ML(Q) ensures quality at every stage of the ML workflow. MLOps creates reproducible, production-ready pipelines, automating and optimizing model development, training, deployment, and monitoring. CRISP-ML(Q) guides each step of an ML project through six phases, as shown in Fig. 2 [14]. Together, MLOps and CRISP-ML(Q) enable the shift from one-off experiments to traceable, continuously improving ML systems ready for clinical deployment. This integration facilitates the development of high-quality models capable of supporting real-time seizure prediction.

4.1. Business and data understanding

4.1.1. Define business objectives

The goal is to create a system for accurate automated epilepsy prediction while using an MLOps framework that can adapt to specific needs.

4.1.2. Translating business objectives to ML objectives

To achieve our goal, several tasks must be implemented. This involves the following steps:

- Developing a patient-specific model: Build a model that adapts to individual predictions, considering patient variability.
- Identifying key performance indicators (KPIs): Achieve high performance in classifying preictal and interictal, targeting an accuracy of 99% with minimal false alarms.
- Ensuring real-time predictions: Enable the model to predict seizures with minimal inference time for early intervention.
- Implementing scalability and reliability: Build robust ML pipelines for production, including continuous monitoring, feedback, retraining, and performance evaluation.

These objectives align with MLOps principles by promoting continuous delivery, operational stability, and adaptability in dynamic healthcare environments.

4.1.3. Collect and verify data

We used the open-source CHB-MIT Scalp EEG dataset from PhysioNet [49], which is widely employed for epilepsy diagnosis. It comprises EEG data from 24 pediatric epilepsy patients (19 males and 4 females), aged between 2 and 22 years. The dataset contains a total of 1136 h of signal, including 89 seizures that have been manually annotated. EEG activity was captured using 23 electrodes positioned according to the international 10–20 system, sampled at 256 Hz, and stored in EDF (European Data Format) files. Each file includes precise annotations of seizure onset.

4.1.4. Assess the project feasibility

This task evaluates key factors critical to the project's success:

- Data quality and availability: Noise and artifacts, such as eye blinks, must be removed.
- Infrastructure and scalability: The need for low-latency architectures, such as edge computing or optimized cloud setups.
- Model generalization: Avoid overfitting, where a model works well during training but poorly on new data.
- Automation and challenges: Reduce complexity in developing automated pipelines due to challenges related to the compatibility of MLOps tools and environment scalability.

4.2. Data engineering

4.2.1. Data selection

Several important rules were applied in selecting EEG signals. To improve model performance, only seizures with sufficient preictal duration were included. Patients with a preictal duration of less than 100 min (e.g., ch11 and ch19) were excluded. Also, patients with changes in their EEG recording setup (e.g., ch12 and ch13) were discarded to ensure consistent channel placement across all 23 electrodes. Preictal periods were defined as the 45 min before a seizure. If there were consecutive seizures, a 30-minute period after the first seizure was excluded to prevent overlap between preictal and postictal states, thus maintaining clear class separation. Finally, we removed ch24 because we defined the interictal period as the window from 4 h before to 1 h after the seizure. This choice was made since there is no clear definition of preictal and postictal phases.

4.2.2. Data cleaning: Noise reduction and artifact removal

EEG signals are complex, nonlinear, and often contaminated by artifacts from eye movements, muscle activity, and power line interference. To enhance signal quality and improve the signal-to-noise ratio, we applied a sequence of filters: a notch filter to remove power line noise, a Butterworth bandpass filter to isolate relevant frequencies, and a Finite Impulse Response (FIR) filter for stability and linear phase response [33,34].

4.2.3. Feature engineering and selection

In addition to deep learning automated feature extraction, we also applied manual extraction for three main reasons [6,33]:

- The MLOps data engineering team uses EEG features to check data quality, integrity, and reliability. These features, summarized in Table 2, help confirm that data follow expected patterns and contain no significant artifacts or anomalies.
- These features improve preprocessing within the MLOps workflow. They boost explainability, reproducibility, and traceability, particularly by allowing real-time inspection of signal quality metrics.
- Feature selection identifies the most relevant EEG channels for classification, reducing model complexity and computational cost by limiting the number of inputs to the most informative channels.

All features listed in Table 2 are automatically computed using Python during preprocessing. Mean, variance, skewness, and kurtosis were calculated for each EEG channel using NumPy and SciPy to summarize signal distribution and detect abnormalities. MSE and RMSE are computed between raw and filtered signals to assess filtering quality, while SNR estimates the strength of the signal relative to noise. Entropy (measures signal complexity). Finally, Fisher Discriminant Ratio and entropy difference are applied to labeled preictal and interictal segments to select the most relevant channels. All computations are embedded in the reusable orchestrated tool and automatically tracked by the experiment tracking.

4.2.4. Channel optimization and selection

Using fewer EEG channels simplified preprocessing and training, especially with wearable devices [7,50]. We selected two EEG channels that yield the best prediction performance based on a combination of SNR, Fisher Discriminant Ratio, and entropy difference between interictal and preictal states listed in Table 2.

4.2.5. Data segmentation and labeling

Breaking down long EEG recordings into smaller time windows helps capture brain states and patterns. Shorter segments reveal spatial details but lack time features, while longer segments show broader patterns but may include noise [51]. We used EEG signals with the raw time-series dimension for our model input. This allowed it to learn directly from raw data to recognize temporal brain signal changes. We also labeled preictal segments as class 1 and interictal segments as class 0. In addition, we tested various segment lengths (5 s, 10 s, 20 s, and 30 s) to find the most effective window duration.

4.2.6. Data augmentation, class balancing, and standardization

To handle imbalanced datasets and avoid bias, we used overlapping time windows (50% to 83.33%, 2 to 6 ratio) to increase the number of preictal segments. We selected the best segmentation settings based on its performance. For 1:1 balanced classes, we randomly selected interictal segments and evenly distributed them across EEG files to minimize bias [51]. Additionally, we standardized features using scikit-learn's StandardScaler to ensure a mean of 0 and a standard deviation of 1. For multi-class classification, we prepared labels using one-hot encoding, allowing the model to effectively output class probabilities.

4.2.7. Document the data engineering phase

MLOps facilitate rapid experimentation and model traceability. Previous processing steps are automatically logged. By using an experiment tracking tool to monitor runs-configuration with 2 different high-cut frequencies (40 Hz or 70 Hz). It logs and stores the EDF files, selected best channels, and time-domain features. This MLOps tool makes it easy to track data lineage, compare, and reproduce experiments.

Table 2
Manual feature extraction for EEG analysis with numbered equations.

Feature	Role	Formula (Eq. #)
Mean Squared Error (MSE)	Mean squared difference between raw and filtered signals	$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$ (1)
Root Mean Squared Error (RMSE)	Indicates the magnitude of signal error	$RMSE = \sqrt{MSE}$ (2)
Mean	Average amplitude of the signal	$Mean = \frac{1}{N} \sum_{i=1}^N x_i$ (3)
Variance	Measures signal variability	$Variance = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$ (4)
Skewness	Indicates distribution asymmetry	$Skewness = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^3$ (5)
Kurtosis	Measures signal peakedness	$Kurtosis = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^4 - 3$ (6)
Entropy	Quantifies signal complexity	$H(X) = - \sum p(x_i) \log p(x_i)$ (7)
Signal-to-Noise Ratio (SNR)	Assesses signal vs. noise power	$SNR = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$ (8)
Fisher Discriminant Ratio (FDR)	Separability between interictal and preictal	$FDR = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$ (9)
Entropy difference	Entropy shift between brain states	$\Delta H = H_{preictal} - H_{interictal}$ (10)

4.3. Model engineering

Model engineering involves selecting a baseline, fine-tuning with domain knowledge, and training the final model. The process should be well-documented. Data scientists handle model development and optimization, while MLOps manages infrastructure, streamlines pipelines, and reproduces results. Collaboration ensures efficiency and quality, bridging the gap between prototyping and deployment, aligning with clinical needs.

4.3.1. Model selection

In our work, we chose to develop a CNN-BiLSTM with an attention mechanism [52]. This architecture has a strong ability to handle both spatial and temporal features. The 1D CNN extracts spatial patterns across the best channels, while the Bi-LSTM captures long-term temporal dependencies. The attention mechanism further focuses on the most relevant parts to prioritize informative sequences. Our model architecture is designed to minimize processing steps, using the same format for the EEG signal as input.

4.3.2. Model architecture

This task explains each layer in our model [40]. Table 3 summarizes our architecture layers as follows:

- Input Layer:

This input layer sets the data shape for the model. It processes EEG segments with samples and channels.

- CNN block:

We used four sequential 1D convolutional layers. Each layer applied a convolution block on the input. Each convolutional layer was followed by batch normalization to improve stability and accelerate training convergence. Max pooling was then used to select the maximum value from a fixed-size window, retaining important features and reducing dimensionality.

- LSTM block:

A bidirectional LSTM layer followed the CNN blocks. It learns from both past and future patterns. An LSTM cell includes gates that control the flow of information: the forget gate discards irrelevant details, the input gate determines what new information to store, and the output gate passes relevant information to the next layer. By combining hidden states from both directions, the bidirectional mechanism captures a more comprehensive temporal context. We added dropout layers after each LSTM layer to prevent overfitting by randomly deactivating neurons during training.

- Attention Mechanism:

The attention mechanism computes a weighted sum of LSTM outputs, focusing on the most relevant features. It calculates similarity scores between the query and key vectors. These scores are passed through a softmax function to produce attention weights. Inputs with higher relevance receive greater attention. The final output is computed by applying these weights to the value vectors.

- Flatten and Dense Layers:

The attention outputs are flattened into a 1D vector, enabling dense layers to map features to the final output classes. Dense layers summarize the learned representations for classification. We included two dense layers with ReLU activation to reduce the risk of vanishing gradients during backpropagation.

- Output Layer:

The final layer provides class probabilities. We used the softmax function, which is suitable for both multi-class and binary classification. It standardizes the outputs within the [0, 1] range and ensures that the probabilities sum to 1, producing interpretable confidence scores for each class.

4.3.3. Defining quality measures

We evaluated our model using the precision and robustness metrics, as detailed in Section 4.4. Table 5 summarizes all key metrics used for evaluation, including their mathematical formulas.

We used a confusion matrix and ROC curves. The matrix shows true and false positives and negatives. ROC curves highlight the balance between true and false positive rates. A good model has a curve near the top-left, showing high sensitivity and low false positives.

4.3.4. Model training

Our training process used Stratified k-Fold Cross-Validation (CV) with hold-out, as described in Fig. 3 [53]. We split our dataset into three sets: training, validation, and testing. Hold-out validation extracted 10% of unseen data for final testing.

The remaining 90% were used for Stratified k-Fold CV, which provided training and validation splits. Stratified CV is a variant of k-Fold CV suitable for imbalanced datasets, where each fold maintains the same class distribution as the full dataset. To ensure efficient training within each CV fold, we applied the following techniques:

- *Training from Scratch:* For every CV fold, we retrained the model from scratch, ensuring no information from previous folds was retained.
- *Early Stopping:* We stopped training when validation accuracy no longer improved, preventing overfitting.

Table 3
Model's architecture and parameters.

Layer	Output shape	Param #	Description
Input layer	(None, T , 2)	0	Input time-series EEG data T
Conv1D	(None, $T-2$, 32)	$2 \times (3 \times 2 \times 32 + 32) = 224$	Conv: 32 filters and kernel-size = 3
BatchNormalization	(None, $T-2$, 32)	$2 \times 32 = 128$	Normalization over 32 features
MaxPooling1D	(None, $\frac{T-2}{2}$, 32)	0	Downsample with pool size 2
Conv1D_1	(None, $\frac{T-4}{2}$, 64)	$2 \times (3 \times 32 \times 64 + 64) = 6208$	Convolution with 64 filters
BatchNormalization_1	(None, $\frac{T-4}{2}$, 64)	$2 \times 64 = 256$	Normalization over 64 features
MaxPooling1D_1	(None, $\frac{T-4}{4}$, 64)	0	Downsample with pool size 2
Conv1D_2	(None, $\frac{T-6}{4}$, 128)	$2 \times (3 \times 64 \times 128 + 128) = 24,704$	Convolution with 128 filters
BatchNormalization_2	(None, $\frac{T-6}{4}$, 128)	$2 \times 128 = 512$	Normalization over 128 features
MaxPooling1D_2	(None, $\frac{T-6}{8}$, 128)	0	Downsample with pool size 2
Conv1D_3	(None, $\frac{T-8}{8}$, 256)	$2 \times (3 \times 128 \times 256 + 256) = 98,560$	Convolution with 256 filters
BatchNormalization_3	(None, $\frac{T-8}{8}$, 256)	$2 \times 256 = 1024$	Normalization over 256 features
MaxPooling1D_3	(None, $\frac{T-8}{16}$, 256)	0	Downsample with pool size 2
Bidirectional (LSTM)	(None, $\frac{T-8}{16}$, 64)	$4 \times [64 \times (64 + 256 + 1)] = 73,984$	Bidirectional LSTM with 64 units (32 bidirectional)
Dropout	(None, $\frac{T-8}{16}$, 64)	0	Dropout for regularization
Bidirectional_1 (LSTM)	(None, $\frac{T-8}{16}$, 32)	$4 \times [32 \times (32 + 64 + 1)] = 10,368$	Bidirectional LSTM with 32 (16 bidirectional) units
Dropout_1	(None, $\frac{T-8}{16}$, 32)	0	Dropout for regularization
Attention Mechanism	(None, $\frac{T-8}{16}$, 32)	0	Attention layer over time steps
Flatten	(None, $32 \times \frac{T-8}{16}$)	0	Flatten features
Dense	(None, 128)	$32 \times \frac{T-8}{16} \times 128 + 128$	Fully connected with 128 units
Dropout_2	(None, 128)	0	Dropout for regularization
Dense_1	(None, 64)	$128 \times 64 + 64 = 8256$	Fully connected layer with 64 units
Dropout_3	(None, 64)	0	Dropout for regularization
Dense_2	(None, 2)	$64 \times 2 + 2 = 130$	Fully connected for classification
Total parameters		Dependent on T	Sum of total trainable and non-trainable parameters

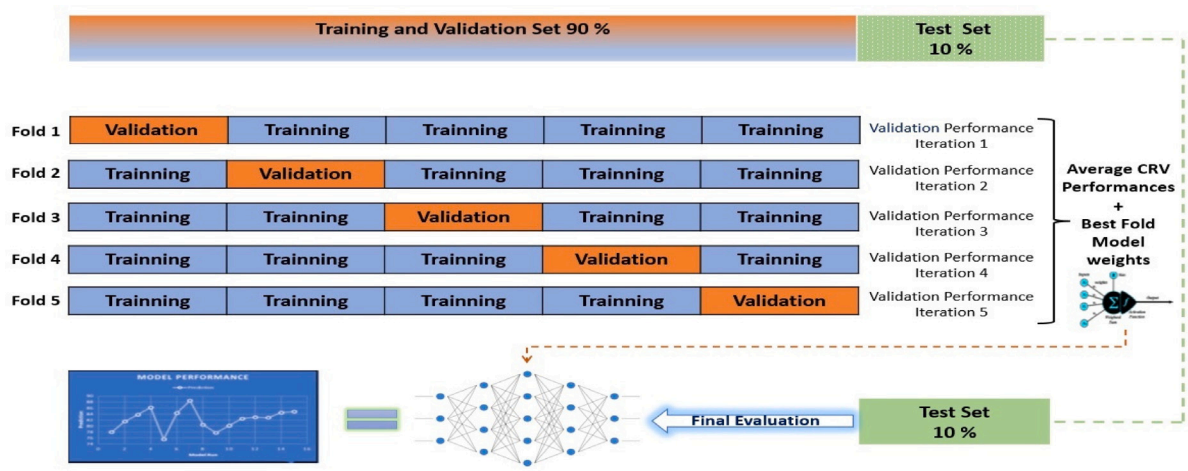


Fig. 3. Training strategy combining a 90% training/validation split and a 10% hold-out test set. The training/validation portion is processed using Stratified 5-Fold Cross-Validation, ensuring balanced class distributions across all folds.

- **Model Checkpointing:** We saved the best-performing model during training based on validation accuracy.
- **StopAtValAccuracy:** A custom callback stopped training when validation accuracy reached 99%.
- **ReduceLROnPlateau:** We reduced the learning rate when the validation loss plateaued, refining the model's performance.
- **Epochs and Batch Size:** We used a maximum of 60 epochs and a batch size of 42 to ensure the model learned effectively without overfitting or underfitting.

4.3.5. Model fine-tuning

We tuned our CNN-BiLSTM model using the Optuna hyperparameter optimization framework [54], as summarized in Table 4. Optuna

facilitates the identification of the most effective configurations, enhancing model precision and reducing training time. Optuna works by defining a search space for the hyperparameters (e.g., kernel size, number of CNN layers, LSTM units, dropout, batch size, and epochs), then evaluating different configurations over multiple trials using intelligent sampling strategies. Each trial is assessed using cross-validated accuracy as the objective metric, and the configuration yielding the highest average accuracy is retained as optimal [55]. These optimized settings were then integrated into the MLOps development pipeline. By reusing the selected parameters across all patient-specific models, we minimized the need for individual re-tuning, thereby improving both the efficiency and consistency of the pipeline.

Table 4
Hyperparameter-tuning results using Optuna.

Parameters	Values/Type
Input layer shape	(None, samples per segment, 2 channels)
Number of Conv1D Layers	4
Number of filters in Conv1D	32, 64, 128, 256
Kernel size in Conv1D	3
Pooling size (MaxPooling1D)	2
Batch normalization	After each Conv1D layer
Number of Bidirectional LSTM Layers	2
LSTM units	32 (first layer), 16 (second layer)
Attention mechanism	Applied to LSTM output
Dropout rate	0.1 for LSTM layers, 0.2 for Dense layers
Flatten layer	After attention mechanism
Number of dense layers	2
Dense layer units	128, 64
Activation function (Dense layers)	ReLU
Output layer units	2 (binary classification)
Activation function (Output Layer)	Softmax
Optimizer	Adam
Learning rate	0.001
Loss function	Categorical Crossentropy

4.3.6. Documenting the developed model

MLOps supports model explainability integration at every stage. We used experiment tracking to document each stage of model development. Every experiment consisted of a collection of runs, each representing a single execution of training. For each run, we tracked artifacts, configurations, and hyperparameters. K-fold cross-validation results were grouped under a parent run, and the best-performing checkpoint was saved. We also recorded detailed dataset information, including size, selected channels, and extracted features, to ensure full reproducibility. This supports auditability and ensures compliance with clinical and regulatory standards.

4.4. Model evaluation

This phase assesses whether trained models meet deployment criteria. A results, metrics from Table 5 are computed programmatically during both cross validation training and testing. After each fold, accuracy, precision, recall, F1-score, specificity, FPR, and FPH are calculated using the confusion matrix and stored. The results are averaged across all folds to get the final cross-validation performance. In testing, the trained model is applied to unseen patient data, and the same metrics are computed for each patient separately. FPH is calculated by dividing the number of false positives by the total monitoring time in hours. Robustness is measured by running the model on noisy EEG data and comparing the accuracy. All computations use standard Python libraries, and results are logged using an experiment tracker, with per-patient and per-fold metrics saved for full traceability.

4.4.1. Validate model performance

We adjusted key parameters such as high-cut frequency, EEG sampling rate, segment length, and overlap ratio. These dynamics were automatically managed by our MLOps pipeline, ensuring systematic exploration without manual intervention. We trained models using different channel selection strategies (SNR, Fisher Discriminant Ratio, and Entropy Difference) and evaluated their results using accuracy, sensitivity, FPR, and FPH. The pipeline conducted these experiments across all patients and saved the best-performing model.

4.4.2. Decide whether to deploy the model

After the evaluation, we confirmed the model meets our goals: it generalizes well, is robust, and results are explainable. Deployment was decided by comparing experiments and selecting the top configurations tracked in the MLOps pipeline.

4.4.3. Document the evaluation phase

All experiments, including training as well as testing, were carefully tracked. Performance scores were averaged across folds to measure generalization and avoid overfitting risk. Confusion matrices and ROC curves for each fold were captured for increased explainability. Channel selection methods and configuration parameters were logged per run. All models were evaluated on an unseen test set after CV to check for their ability to generalize. Such logs allow for making reliable comparisons before deployment.

4.5. Model deployment

Using a standardized format that works with scalable serving systems, we implemented patient-specific models for real-time inference. To guarantee consistency and manageability across versions, each trained model was exported along with its architecture, weights, and metadata. These models were designed to facilitate efficient inference in production, version tracking, and quick loading.

The deployment workflow treats each patient-specific model as an independent unit, allowing targeted updates and flexible resource allocation. Versioning was applied to manage model evolution over time, enabling rollback or replacement without interrupting service. Models were exposed through web APIs for seamless integration with clinical systems. To ensure functional correctness and deployment readiness, models were validated post-deployment using hold-out datasets. This confirmed that inference behavior in production remained consistent with expected outputs.

4.6. Model monitoring and maintenance in production

MLOps aligns machine learning with enterprise software practices. In production, we continuously monitor deployed models. We tracked key indicators such as health state, requests per second (RPS), inference latency, and error rates. By enabling continuous evaluation and feedback, MLOps minimizes the risk of model degradation and ensures the long-term health of deployed systems. Maintenance involved automated logging, anomaly detection and alerting, and periodic model updates to maintain long-term robustness and scalability.

5. Implementation of MLOps framework

To implement our end-to-end MLOps framework, we integrate several advanced tools, as illustrated in Fig. 4. The entire environment is containerized using Docker, with NVIDIA support, enabling optimized deployment through the cuDNN library [56]. Pipeline orchestration is handled by ZenML, which ensures automation and reproducibility across preprocessing, training, and deployment stages [57]. Storage is managed using MinIO, an S3-compatible system that offers efficient and scalable EEG data handling [58].

Model training takes place on Google Colab using T4 High-RAM instances [59], connected to ZenML and MLflow services hosted on-premises. This setup of MLOps transforms experimental notebooks into production-grade workflows and enables seamless integration between cloud-based and local infrastructure [60]. MLflow tracks experiments, including model configurations, hyperparameters, and performance metrics [61,62]. This modular structure reduces technical friction and accelerates the transition from experimentation to production. Once trained, models logged to MLflow are transformed and deployed using TensorFlow Extended Serving [63], which exposes them as REST services suitable for real-time clinical integration. Model inference metrics are monitored using Prometheus and Grafana [64].

Our workflow detailed in algorithm 1 follows the CRISP-ML(Q) methodology and relies on automated pipelines to enforce structure and maintain quality. All tools are integrated into pipelines that span preprocessing, training, deployment, and monitoring. Together, they

Table 5
Performance metrics for model classification.

Metric	Equation	Description
Accuracy	(11) $\frac{TP+TN}{TP+TN+FP+FN}$	Proportion of correct predictions across all instances.
Sensitivity (Recall)	(12) $\frac{TP}{TP+FN}$	Measures the ability to correctly predict true positives.
Specificity	(13) $\frac{TN}{TN+FP}$	Proportion of true negatives correctly identified. Reduces false positives.
Precision	(14) $\frac{TP}{TP+FP}$	Indicates how predicted seizures result in actual seizures.
F1-Score	(15) $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall. Useful in imbalanced datasets.
FPR	(16) $\frac{FP}{FP+TN}$	Frequency of incorrect preictal predictions when no seizure occurs.
FPH	(17) $\frac{FP}{T_{\text{Monitor}}}$	Frequency of false positives per hour during monitoring. Important in real-time systems.
Robustness	Accuracy with noisy data	Model's ability to maintain performance under noisy data.

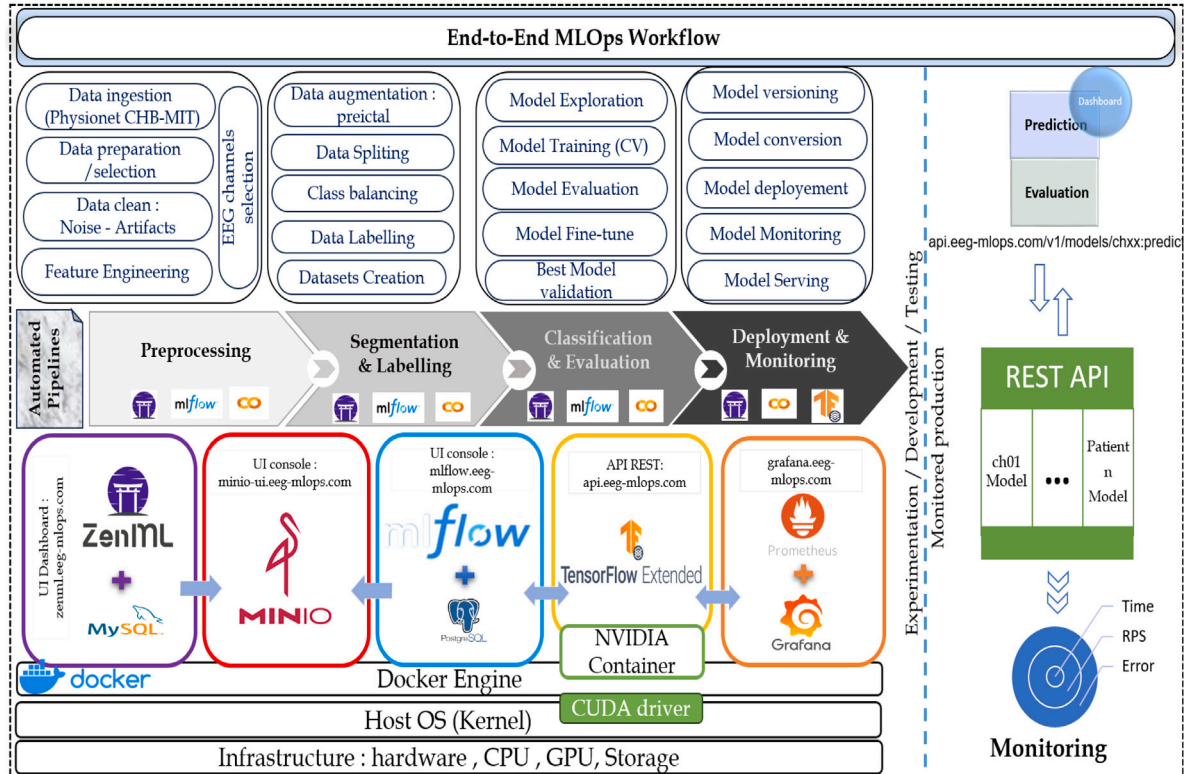


Fig. 4. End-to-End MLOps Workflow for EEG Seizure Prediction. Stacks, tools, pipelines, and steps used. The MLOps stack provides tools for versioning, orchestration, deployment, and monitoring.

provide a framework that guarantees operational discipline while preserving scientific rigor and model quality. The preprocessing pipeline starts with cleaning and preparing EEG data. ZenML manages steps from data ingestion to feature extraction. It applies noise removal and channel selection to enhance data quality. Configurations and features are logged in MLFlow for artifact lineage and dataset versioning. Cleaned EEG data is then segmented and labeled by ZenML. Segments are marked as preictal or interictal, with details tracked in MLFlow. These datasets are reused in training and evaluation.

Next, patient-specific models were trained with segmented data. ZenML handles the training pipeline and data transfer to deep learning frameworks. Training details and metrics are logged in MLFlow. Models are validated via five-fold cross-validation, with the top config chosen automatically. This workflow ensures reproducibility and adaptability across patients or institutions. After training, top models are packaged with TensorFlow Extended. Versions are logged for traceability and reproducibility. Patient-specific models are deployed as REST APIs, containerized with Docker and NVIDIA support. Multiple models run simultaneously for real-time inference.

In production, a real-time dashboard shows alerts for consecutive preictal segments. Predicted labels are checked against true data if

available. Inference quality is tracked with confusion matrices and metrics. Real-time predictions are monitored for speed, reliability, and accuracy to reduce false alarms and support early clinical intervention. Prometheus tracks key inference metrics and links to Grafana for live model performance visuals and alerts. These tools support scalable, reliable seizure prediction models. They enable modular, reusable pipeline components for better clinical care.

6. Results and discussion

This section explains the main results from our automated system. We look at how well the model works in different settings, how it handles new data, and the importance of each component of the MLOps system in making it reliable, scalable, and practical in the real healthcare environment for epilepsy diagnosis.

6.1. Analysis of EEG channel selection

Fig. 5 illustrates the best EEG channels selected across the preprocessing phase. The FZ-CZ channel is the most chosen. It was selected 15

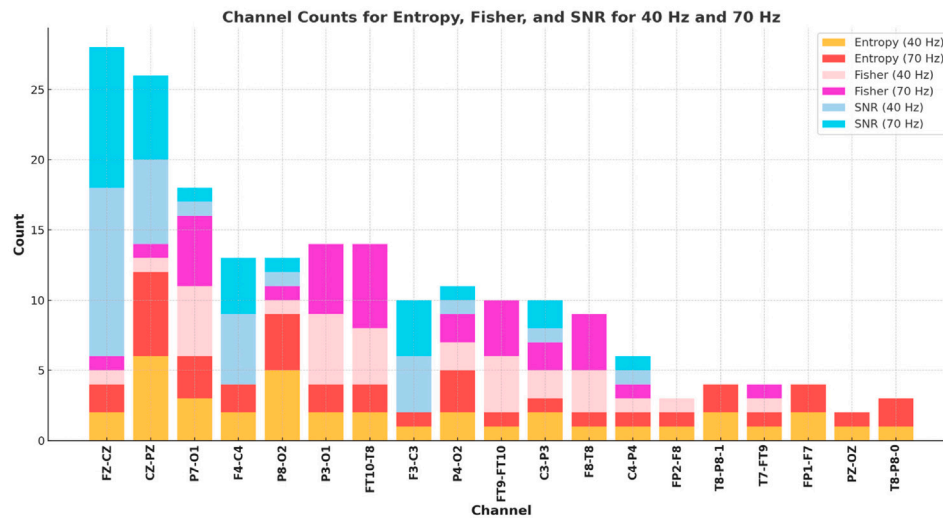


Fig. 5. Comparison of channel selection counts for Entropy, Fisher, and SNR features at 40 Hz and 70 Hz. Channels are labeled according to the international 10–20 EEG system, where: F = Frontal, P = Parietal, C = Central, T = Temporal, O = Occipital. Each label represents a bipolar montage.

times at 40 Hz and 13 times at 70 Hz, particularly for the SNR feature (12 times at 40 Hz and 10 times at 70 Hz). It produces signals with low noise but is less effective for entropy and Fisher. CZ-PZ was the second-best channel. It appeared 13 times for 40 Hz and 12 times for 70 Hz, mainly for entropy and SNR (6 times each), while Fisher had only one selection. P7-O1 was the most selected channel using Fisher (6 times for both frequencies) but showed fewer selections for SNR. Channels like F4-C4 and P8-O2 performed well for entropy and SNR but had low prominence for Fisher, indicating good signal quality but limited separability. P7-O1, P3-O1, and FT10-T8 were balanced across features.

In general, frontal and central channels like FZ-CZ and CZ-PZ are frequently selected using SNR. This suggests these regions are less affected by noise. Fisher favors P7-O1, highlighting its role in distinguishing preictal from interictal phases. Entropy selections were more evenly distributed across brain regions due to variations in seizure onset type and location among patients.

This analysis highlights the importance of automating the channel selection process. The MLOps workflow makes signal quality assessment more efficient with the choice of the best brain region. Also, feature-driven channel selection allows for comparing results across different patients, which can support future studies on region-specific pre-seizure biomarkers.

6.2. Performance evaluation of the developed model

We computed the average performance across 5-fold CV to identify the best model configurations, which include two high-cut frequencies, three channel selection features, four segment lengths, and five overlap levels. In total, 3420 runs were tracked in MLFlow, enabling transparent and reproducible comparison of all experimental results. This large-scale benchmarking of configuration allowed for data-driven setting discovery across patients.

6.2.1. Performance evaluation across patients

Fig. 6 presents a grouped boxplot generated using MLFlow, summarizing the distribution of model metrics across all patients. Boxplots provide a general impression of the model performance variability across patients. A box indicates the distribution of a specific measure (e.g., accuracy, sensitivity) for all patient configurations. The central line indicates the median, the box covers the interquartile range (IQR),

and whiskers extend to 1.5. Single dots represent outliers, which indicate very strong or very weak configurations. A patient performs stably and well when the box is small, the median is high (close to 1.0 for precision or close to 0 for false positives), and there are no or few outliers that result in robustness to changes in settings. Large boxes with lots of outliers indicate that performance is extremely variable and configuration-sensitive. Low-medians, highly variable patients may require sensitive tuning.

Patients like ch01, ch03, and ch23 achieved outstanding accuracy, F1-score, precision, sensitivity, and specificity, with median values around 99% and small IQRs. This indicates stable and strong performance across configurations. In contrast, patients like ch02, ch04, ch07, and ch22 had wider IQRs, more outliers, and lower metrics. This highlights configuration-dependent performance and challenges in balancing true and false positives. Adjusting configurations for these patients is mandatory to increase metrics performance. Patients such as ch09, ch15, and ch20 showed good overall performance (sensitivity = 0.949 and specificity = 0.938) but with some variation, demonstrating the model's fine-tuning potential.

Additionally, ch01 had low median FPH (2.17) and FPR (0.0069), indicating minimal false alarms. However, ch04 and ch07 had high FPH (30.76, 36.71) and FPR (0.1052, 0.1509), reflecting difficulties in distinguishing interictal from preictal and complex data. Moreover, average training time also varied with patients. For example, ch06 required 3200 s, while ch15 took 2500 s. Patients such as ch01 and ch23 required less time (500–800 s), likely due to fewer EDF files and simpler EEG signals. For instance, ch06 had 18 EDF files with 10 seizures, while ch01 had 33 EDF files with only 4 seizures.

Overall, every patient-specific model demonstrated strong performance, even when some showed high variability. Despite wide IQRs in certain cases, the best configuration still achieves high accuracy. These results highlight the importance of automated identification of optimal settings, such as high-cut frequency, segment length, and overlap ratio, which will be explored in detail in the following sections.

6.2.2. Performance evaluation across pass-band filters

Fig. 7 shows boxplots of model metrics across two high-cut frequencies (40 Hz and 70 Hz). At 70 Hz, most metrics showed better results than at 40 Hz, with higher medians and less variation across configurations. Average accuracy ranged from 0.93 to 0.988, with a median of 0.97. The F1-score was also distributed in the same manner. Precision, sensitivity, and specificity were all higher at 70 Hz. Also,

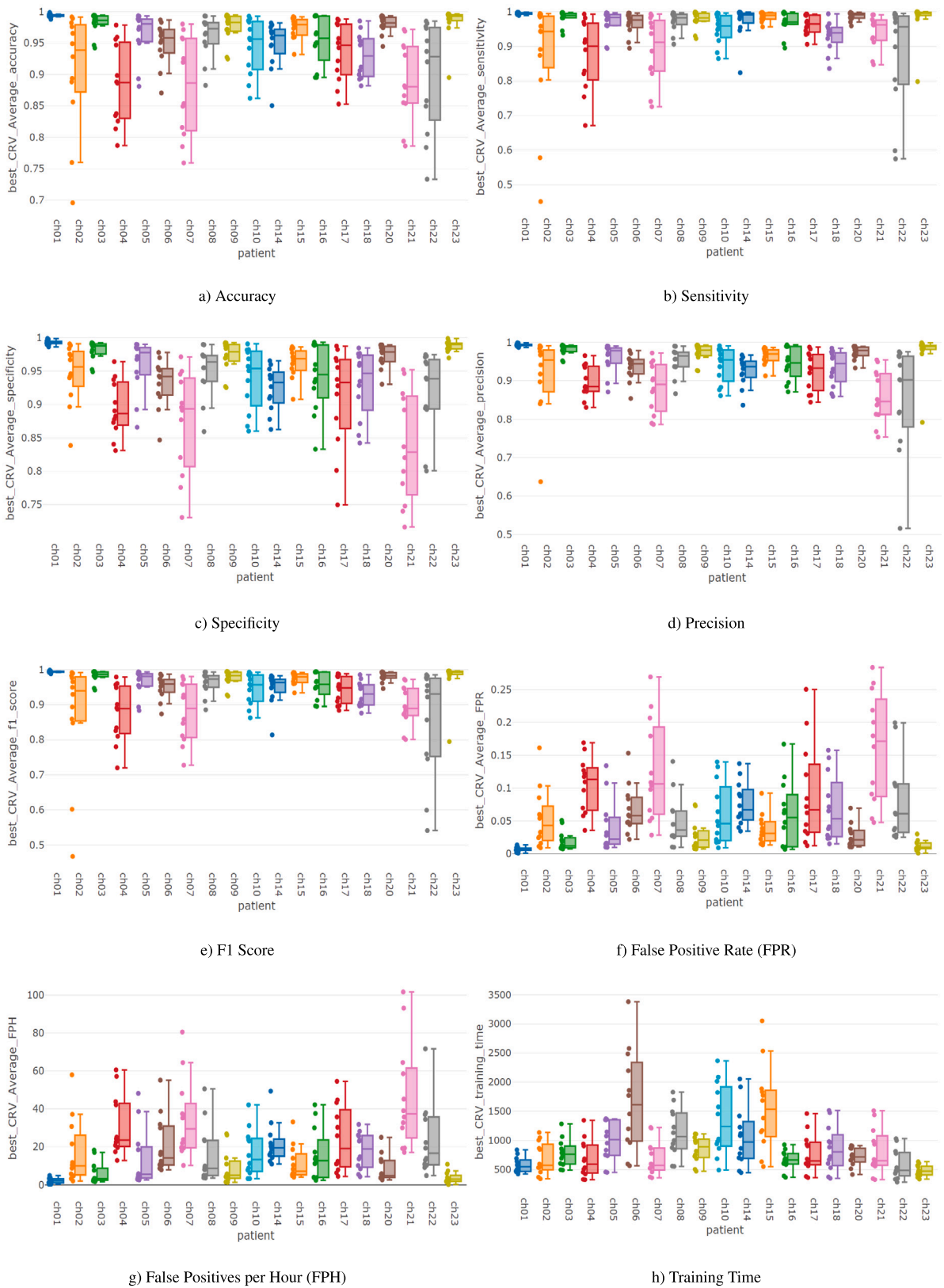


Fig. 6. Boxplots of Model Performance metrics across different patients.

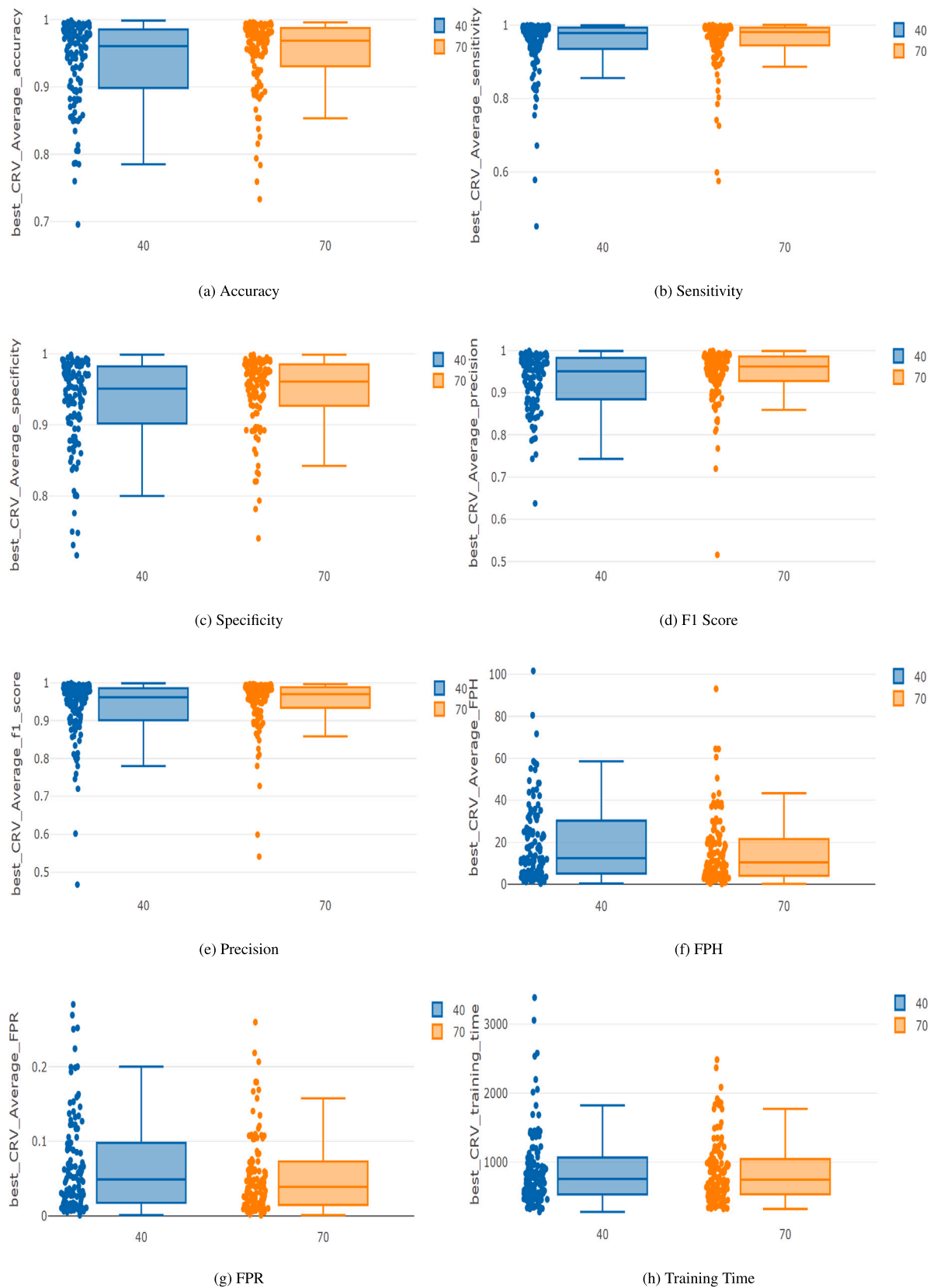


Fig. 7. Boxplots of model performance metrics for different high-cut frequencies (40 Hz vs. 70 Hz) across all patients.

Algorithm 1 Comprehensive step-by-step End-to-End-MLOps-CRISP-ML(Q) workflow for epilepsy prediction

```

1: Input: EEG dataset (CHB-MIT) annotated with seizure onset information
2: Output: Patient-specific CNN-BiLSTM-Attention model for inference in
   real-production environment
3: Main ZenML Orchestrator:
4: for combinations of patients, highcut, segment-lengths, and overlap-ratios
   do
5:   ZenML Pipeline and MLFlow experiment 1: EEG Preprocessing
6:   Step 1.1: Data Acquisition
7:     Load EDF EEG files from MinIO storage with selection criteria,
       and annotate interictal and preictal phases
8:   Step 1.2: Data Preprocessing
9:     For each EDF file, remove invalid channels and apply filters:
       Butterworth, notch, and FIR cleaning
10:    Extract and log signal quality features (MSE, RMSE, mean,
       variance, skewness, kurtosis)
11:   Step 1.3: Channel Selection
12:   for each method (SNR, Fisher Discriminant Ratio, Entropy Difference) do
13:     Compute and log scores for all valid channels. Extract final
       signals based on top-ranked channels
14:   end for
15:   ZenML Pipeline and MLFlow Experiment 2: Segmentation and
     Labeling
16:     Automates segmentation, data augmentation and labeled datasets
       creation
17:   for each method (SNR, Fisher, Entropy) do
18:     Step 2.1: Segmentation and Labeling – Load selected signals,
       extract preictal (label 1) and interictal (label 0) segments using sliding
       and random windows.
19:     Step 2.2: Dataset Construction – Combine, balance, and version
       the labeled segments; store them in MinIO.
20:   end for
21:   ZenML Pipeline and MLflow Experiment 3: Model Training,
     Evaluation, and Selection
22:   Step 3.1: Dataset Splitting and Storage – Split the final dataset into
       90% training/validation and 10% testing; store in versioned format in
       MinIO.
23:   Step 3.2: Model Training – Train a CNN-BiLSTM-Attention model
       with 5-fold cross-validation using segments from SNR, Fisher, and
       Entropy-based datasets.
24:   Step 3.3: Evaluation and Aggregation – Log fold-wise metrics, vi-
       sualize confusion matrices and ROC curves, and compute average CV
       performance.
25:   Step 3.4: Persistence and Registration – Select the highest-
       performing model based on CV accuracy. Save and register it with the
       inferred input signature in MLFlow.
26:   Step 3.5: Final Testing – Evaluate the inferred model on the unseen
       test set and log all metrics and plots.
27:   ZenML Pipeline 4: Deployment and Monitoring
28:   Step 4.1: Model Export and Deployment – Convert the Keras model
       to .pb SavedModel and deploy it in a container.
29:   Step 4.2: API Exposure and Routing – Expose the model via REST
       API and route requests through a secured custom domain.
30:   Step 4.3: Monitoring and Alerting – Connect all models container to
       Prometheus-Grafana to track inference metrics (latency, RPS, queue), and
       trigger alerts in case of drift or performance degradation.
31: end for

```

models at 70 Hz produced fewer false positives and false negatives. These results are in line with the view that using a higher cutoff frequency, particularly within the gamma band, is able to detect important preictal patterns such as sharp waves, spikes, and high-frequency oscillations that often accompany higher seizure risk.

Although certain research points out the importance of frequencies below 35 Hz, we determine from our results that upper bands are dense in predictive information. Therefore, we do not hard-code a fixed 0–70 Hz frequency for each patient. Our pipeline includes automated selection. The MLOps pipeline experiments select the optimal frequency

per patient. This offers personalized processing that adapts to the patient's own characteristics and maximizes performance.

6.2.3. Performance evaluation across channel selection

Boxplots in Fig. 8 show the performance of channel selection methods. Fisher performed best, with a median accuracy of 0.975 and a F1-score of 0.976. SNR-based channels achieved an accuracy of 0.95, similar to entropy, which exhibited greater variability. Fisher also provided the best balance between sensitivity and specificity, at 0.98 and 0.969 respectively. Regarding false positives, Fisher channels achieved the lowest FPR of 0.03 and FPH of 10. Entropy channels followed with an FPR of 0.05 and FPH of 11, while SNR channels had the highest FPH of 17. In terms of training time, Fisher-based selections were the fastest, averaging 587 s. SNR-based channels required 789 s, and entropy-based channels took the longest at 975 s.

In summary, Fisher enhances separability between preictal and interictal phases, resulting in accurate and faster classification during training. SNR and entropy-based selections also yielded good performance, confirming the effectiveness of our MLOps preprocessing strategies to detect the most relevant brain regions.

6.2.4. Performance evaluation across segments and overlaps

Table 6 illustrates the effect of overlap-based data augmentation on the number of 30-second preictal segments. By applying an overlap ratio of 1/6 (stride = 5 s), we consistently observe a sixfold increase in segment count across all patients. The same number of interictal segments were used, effectively balancing classes and doubling the dataset size. On average, the overlapping configuration yields 2383 preictal segments per patient, compared to only 405 without overlap. This increase is visually depicted in Fig. 9-d. Interestingly, other configurations—such as 5-second windows with no overlap, 10-second windows with 1/2 overlap, and 20-second windows with 1/4 overlap, generate a comparable number of segments, offering flexibility in data design. These design choices significantly impact model performance, as illustrated in Fig. 9. Short segments (e.g., 5 s) provide a high dataset size but fail to capture sufficient context, resulting in modest accuracy and F1-scores around 0.90 and increased false positives (FPR = 0.11, FPH = 39.95). In contrast, longer segments offer broader temporal information that better aligns with preictal evolution. The best results of accuracy and F1-score near 0.99, sensitivity at 0.99, and specificity at 0.98, were achieved using 30-second segments with 1/6 overlap, where FPR and FPH dropped to 0.02 and 7.09, respectively. However, these gains come with increased computational cost: the training time rises from 439 s (5 s windows) to over 1400 s with 30 s windows. This trade-off highlights the importance of synchronizing the input structure with the model architecture. LSTM layers benefit from extended temporal context, requiring enough timesteps to model long-range dependencies effectively. The attention mechanism further amplifies this by focusing on informative subsequences within longer windows. With richer sequences, it can better identify seizure-relevant dynamics. Such nuanced optimization would be infeasible to perform manually at scale. Here, MLOps play a decisive role. Rather than relying on static heuristics, the pipeline systematically benchmarks each configuration, identifies optimal trade-offs, and maintains full reproducibility. This synergy between segment design, model architecture, and automated pipeline execution is key to achieving robust and reliable seizure prediction.

6.2.5. Evaluation of model performance over MLOps workflow

Table 7 and the parallel coordinate plot in Fig. 10 present the best model configurations selected for each patient through our MLOps pipelines. The obtained accuracies ranged from 0.969 to 1.000. Notably, eleven out of nineteen patients achieved accuracy above 0.99, while six others exceeded 0.98. Only one patient (ch21) recorded an accuracy slightly below 0.98 (0.975), highlighting the consistently high classification performance across the cohort. Sensitivity remained

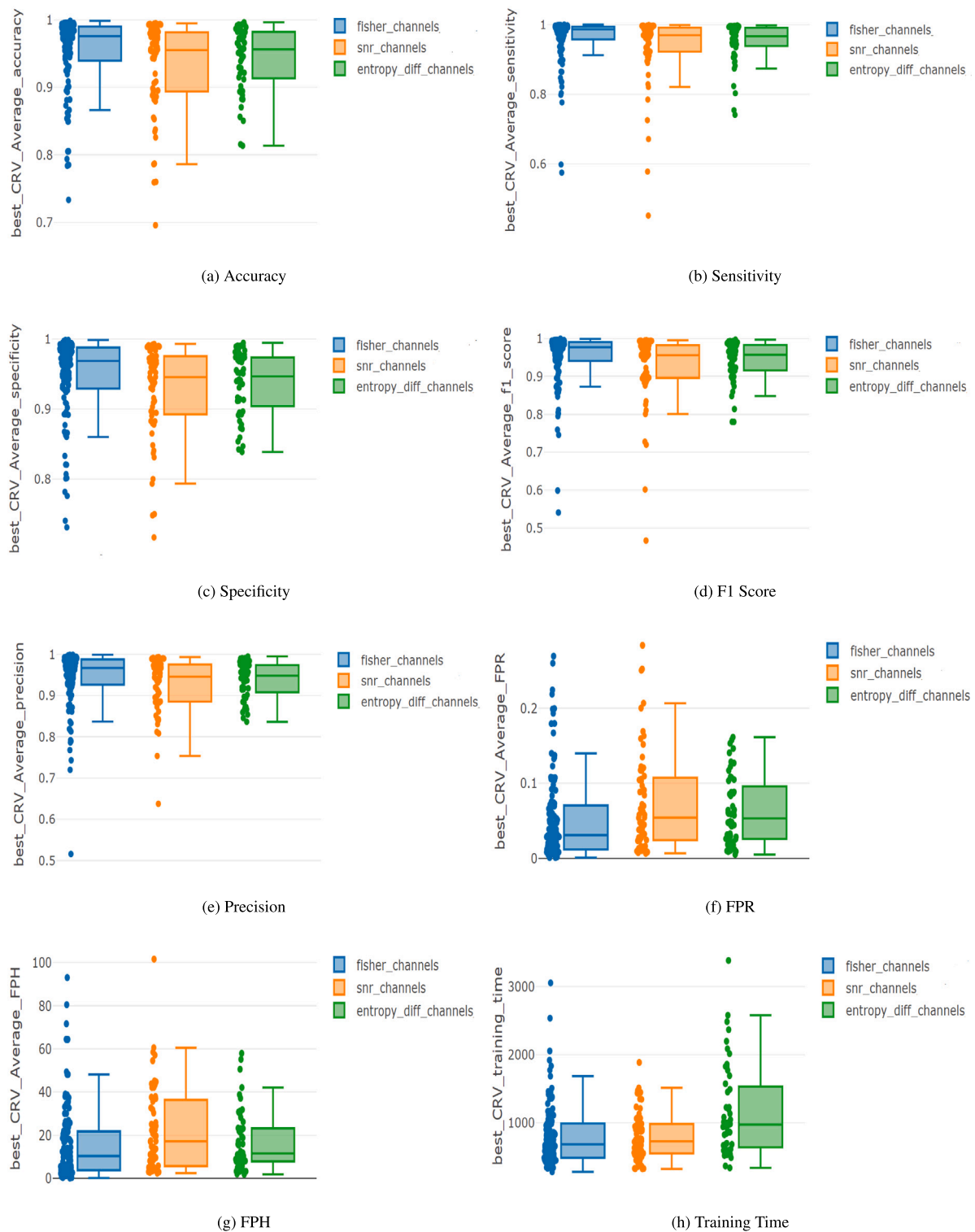


Fig. 8. Boxplots of model performance metrics across different channel selection features (Entropy Difference, Fisher, and SNR).

particularly strong, exceeding 0.993 on average, confirming the model's capacity to reliably detect preictal states.

To assess inter-patient variability in model performance, we applied a Wilcoxon signed-rank test to test accuracies across 19 patients. The results showed no statistically significant difference between patient subsets ($p = 0.1641$), suggesting consistent performance of the proposed model across different individuals. This is further supported by the low standard deviation (0.0083) and high mean accuracy (0.9883), with 75% of patients achieving test accuracy above 0.995.

Patient-specific variability emerged in response to high-cut frequency. Twelve patients achieved superior results with a 70 Hz cutoff, while seven performed better with 40 Hz. This reinforces the importance of tuning parameters to individual EEG spectral characteristics, as seizure morphology and signal-to-noise profiles can vary significantly from one patient to another. All final configurations relied on 30-second window lengths with 5 s strides, which offered the best trade-off between temporal context and data volume. In terms of channel selection, Fisher Discriminant Ratio was the most commonly associated feature

Table 6

Impact of data augmentation on the number of preictal segments per patient. The number of interictal segments is equal to the preictal ones, ensuring class balance. The total dataset size per configuration is therefore the sum of preictal and interictal segments.

Patient	Seizure count	Total preictal duration (min)	Number of preictal segments (30 s)	
			No overlap	Overlap (stride = 5 s)
ch01	7	271.82	523	3138
ch02	3	135.00	270	1600
ch03	7	166.43	332	1960
ch04	3	117.98	235	1400
ch05	6	220.87	440	2600
ch06	7	298.73	595	3573
ch07	3	135.00	270	1605
ch08	5	224.50	446	2657
ch09	4	180.00	360	2140
ch10	7	287.75	574	3417
ch14	7	237.65	473	2809
ch15	12	434.92	869	5117
ch16	4	153.65	307	1806
ch17	3	128.30	256	1521
ch18	4	148.18	296	1757
ch20	6	167.10	334	1973
ch21	4	164.72	329	1949
ch22	3	111.05	222	1317
ch23	4	181.23	362	2149
Average	5.42	202.23	405	2383

Table 7

Patient-wise metrics, configuration, and channel selection.

Patient	Accuracy		Sensitivity		FPR		Config	Best channels
	CV	Test	CV	Test	CV	Test		
ch01	0.999	1.000	0.998	1.000	0.001	0.000	30/6/40	C3-P3, P8-O2 (Fisher)
ch02	0.992	0.987	0.992	0.993	0.009	0.018	30/6/70	F4-C4, FP1-F7 (Entropy)
ch03	0.994	0.997	0.997	1.000	0.009	0.005	30/6/40	CZ-PZ, FZ-CZ (Fisher)
ch04	0.979	0.989	0.993	0.985	0.036	0.007	30/6/70	CZ-PZ, FZ-CZ (SNR)
ch05	0.993	0.987	0.997	0.988	0.010	0.015	30/6/70	F7-T7, FZ-CZ (Fisher)
ch06	0.987	0.983	0.996	0.994	0.022	0.028	30/6/70	F8-T8, P8-O2 (Entropy)
ch07	0.980	0.978	0.989	0.987	0.029	0.031	30/6/70	C3-P3, P7-O1 (Entropy)
ch08	0.993	0.996	0.997	1.000	0.010	0.007	30/6/70	CZ-PZ, F3-C3 (Entropy)
ch09	0.995	0.995	0.997	0.995	0.008	0.004	30/6/40	T7-P7, C4-P4 (Fisher)
ch10	0.993	0.985	0.995	0.991	0.009	0.020	30/6/70	CZ-PZ, P3-O1 (Entropy)
ch14	0.982	0.980	0.998	0.992	0.035	0.032	30/6/40	FP2-F4, P4-O2 (Fisher)
ch15	0.992	0.991	0.997	0.998	0.013	0.015	30/6/70	F4-C4, FZ-CZ (SNR)
ch16	0.994	1.000	0.999	1.000	0.011	0.000	30/6/70	F3-C3, F4-C4 (SNR)
ch17	0.988	0.990	0.989	0.993	0.012	0.013	30/6/70	T7-P7, F7-T7 (SNR)
ch18	0.985	0.969	0.994	0.994	0.024	0.050	30/6/40	FZ-CZ, CZ-PZ (SNR)
ch20	0.993	0.990	0.998	0.994	0.011	0.015	30/6/40	P7-O1, P3-O1 (Fisher)
ch21	0.972	0.982	0.991	0.985	0.035	0.030	30/6/40	FZ-CZ, F3-C3 (Fisher)
ch22	0.985	0.981	0.995	0.984	0.025	0.022	30/6/70	FT9-FT10, FP2-F4 (Fisher)
ch23	0.996	0.998	0.998	1.000	0.005	0.009	30/6/40	FP2-F8, P7-O1 (Fisher)
Avg	0.99	0.989	0.995	0.993	0.017	0.017	30/6/70	FZ-CZ, CZ-PZ

across optimal channels, suggesting its superior ability to capture class separability for most patients. However, optimal channel selection features varied with brain region involvement, confirming the need for adaptive and patient-specific preprocessing. Fig. 11 offers a comparative view between our automated MLOps-driven workflow and static baselines using fixed configurations. Across all performance metrics the MLOps workflow delivers consistently higher results. Unlike static approaches that apply uniform preprocessing and modeling parameters, our adaptive pipeline selects the optimal configuration for each patient through systematic experimentation and automated tracking. Beyond performance, the MLOps infrastructure enables full lifecycle management: from dataset versioning to configuration logging, experiment tracking, and automated model deployment. This ensures not only reproducibility and traceability but also the scalability of the approach for real-time clinical integration.

6.2.6. Performance evaluation of CNN-BiLSTM-Attention architecture

This section compares our model to existing works. As shown in Table 8, our model surpasses both classical ML classifiers (KNN, RF,

SVM) and recent deep learning attention-based approaches. Also, we analyzed the model's progression from a simple 1D CNN to a hybrid CNN-BiLSTM, and finally to our proposed attention-augmented architecture.

For instance, Ma et al. [72] report 94.83% accuracy using channel-level attention, while Quadri et al. [70] achieve 90.54% sensitivity with a DRSN-GRU architecture. In contrast, our MLOps pipeline automatically optimized not only the architecture but also data-driven preprocessing, yielding consistently high scores across all patients, as illustrated in Fig. 10. As shown in the bottom rows from Table 8, the baseline 1D-CNN achieves 84.82% accuracy, 81.43% sensitivity, and 88.03% specificity. Adding BiLSTM layers, which capture bidirectional temporal dependencies in EEG sequences, significantly improves performance boosting accuracy to 90.55% and reducing the false positive rate to 0.108 FPR/h.

To statistically validate the model evolution, we performed a Friedman test on Accuracy, Sensitivity, and Specificity across seven internal models. The results showed a significant performance difference ($\chi^2 = 16.99$, $p = 0.0093$), confirming that the improvements are not due to

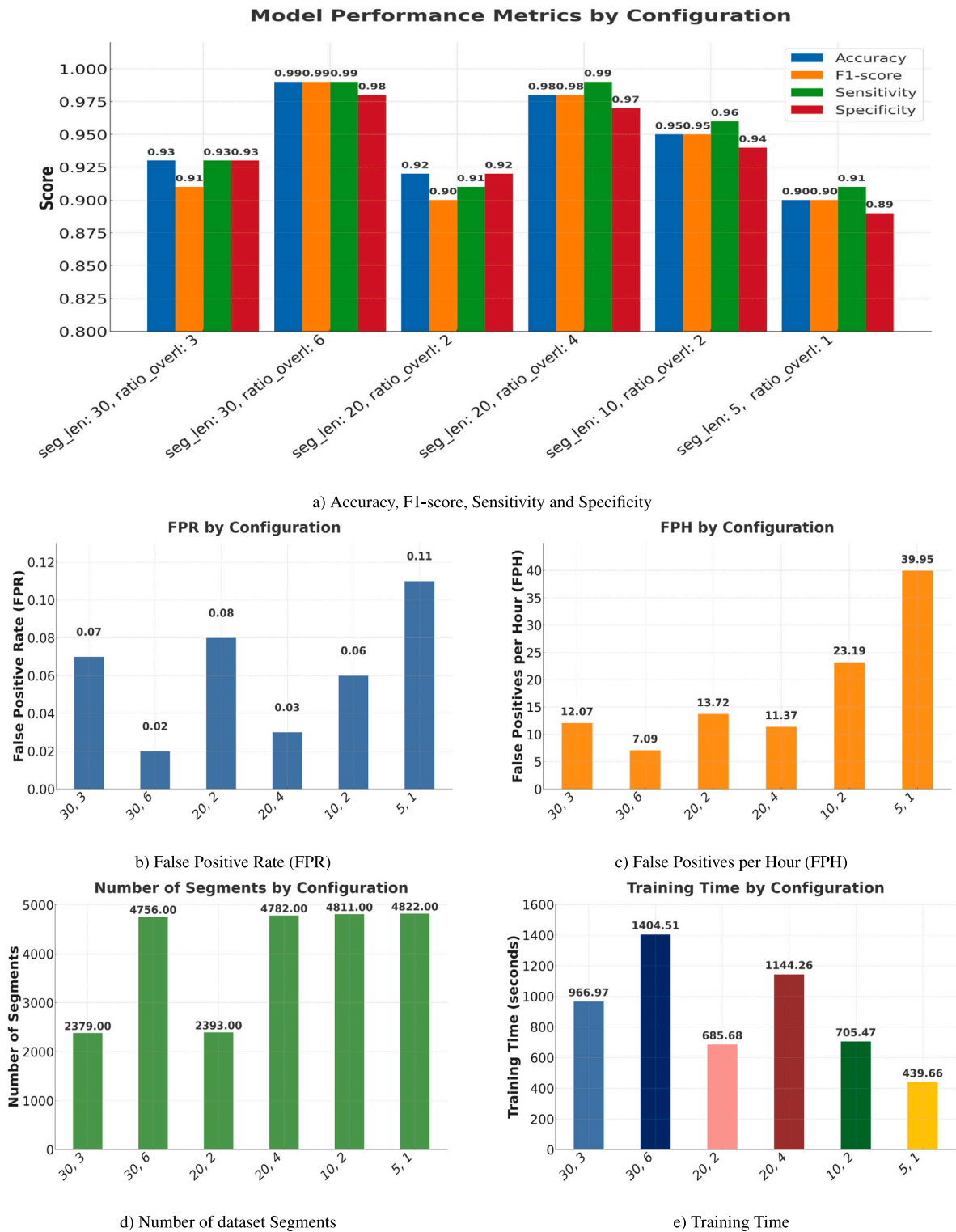


Fig. 9. Barplots of model performance metrics across different segment length and overlap ratio configurations.

random variation. Pairwise Wilcoxon signed-rank tests showed that our proposed model consistently outperformed all earlier versions, although significance was not reached individually ($p = 0.25$), likely due to the limited sample size. These findings highlight the effectiveness of our enhancements, which raised accuracy from 66.75% (KNN) to 99.0%, while also achieving 99.0% sensitivity, 99.0% specificity, and a reduced false positive rate of 0.017/h.

Furthermore, attention mitigates the limitations of classical RNN-based models, which often compress all time steps into a single latent vector. This bottleneck can obscure transient seizure-related patterns. The attention layer instead assigns learnable weights to each timestep, allowing the model to emphasize relevant signal patterns, such as preictal spikes or rhythmic-discharges, while de-emphasizing irrelevant background noise.

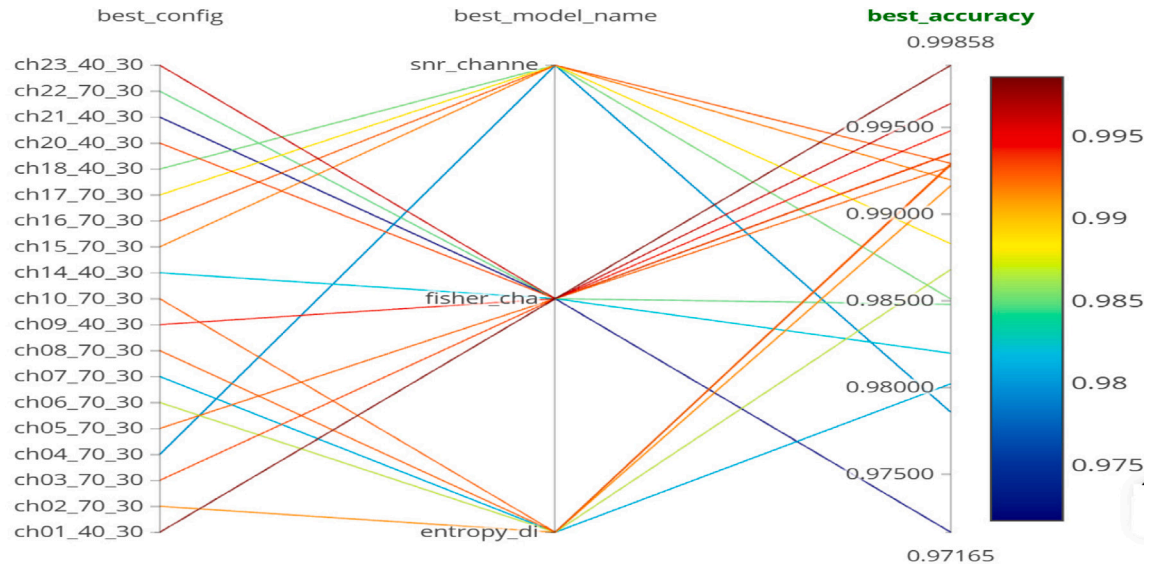


Fig. 10. Parallel coordinates plot illustrating the best model configurations for each patient.

Comparison of MLOps Workflow Performance Across Configurations

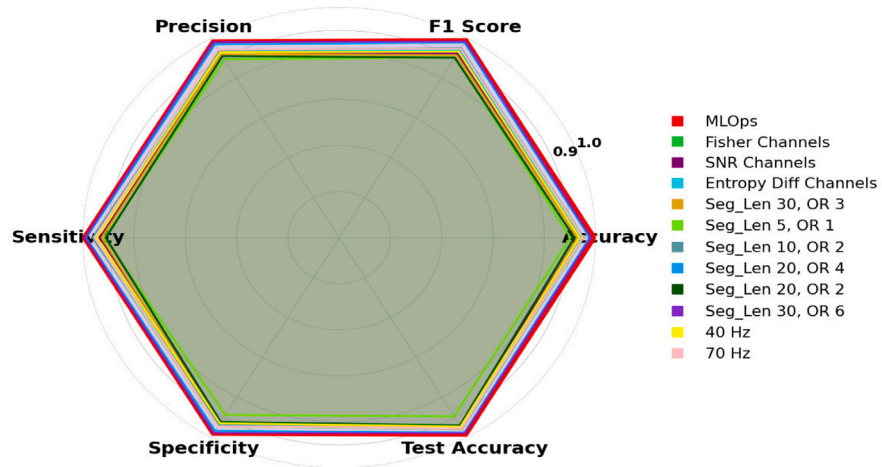


Fig. 11. Radar plots comparing performance across MLOps Workflow versus fixed configurations.

Table 8

Comparison of existing epileptic seizure prediction methods with our proposed method.

Classifier and Method	Accuracy	Sensitivity	Specificity	FPR/h	Prediction horizon (min)
Random Forest with false alarm suppression [65]	–	82.07%	–	0.07	35
CNN + wavelet [66]	88.8%	87.8%	88.3%	0.14	10
CNN+LSTM with GAN augmentation [67]	–	93.0%	92.5%	–	32
CNN/SVM/LSTM with shapelets [68]	–	91.23%	–	0.16	35
Corr.features + shallow CNN [69]	89.98%	92.90%	87.04%	0.13	15
DRSN-GRU with attention [70]	–	90.54%	–	0.11	35
Spiking-GCNN + Federated learning + ANFIS [71]	96.28%	96.33%	96.14%	–	–
CNN-Bi-LSTM with attention-based channel fusion [72]	94.83%	94.84%	94.84%	–	–
1D-CNN + Bi-LSTM with multiscale filters [73]	97.8%	97.63%	–	–	30
Feature-based KNN (our imp)	66.75%	54.76%	77.97%	0.25	45
Feature-based decision tree (our imp)	82.95%	82.19%	83.70%	0.16	45
Feature-based SVM (our imp)	85.88%	84.39%	86.29%	0.137	45
1D CNN (our imp)	84.82%	81.43%	88.03%	0.12	45
LSTM (our imp)	88.38%	88.63%	88.03%	0.145	45
1D-CNN + Bi-LSTM (our imp)	90.55%	90.10%	91.00%	0.10	45
Our proposal	99%	99%	99%	0.017	45

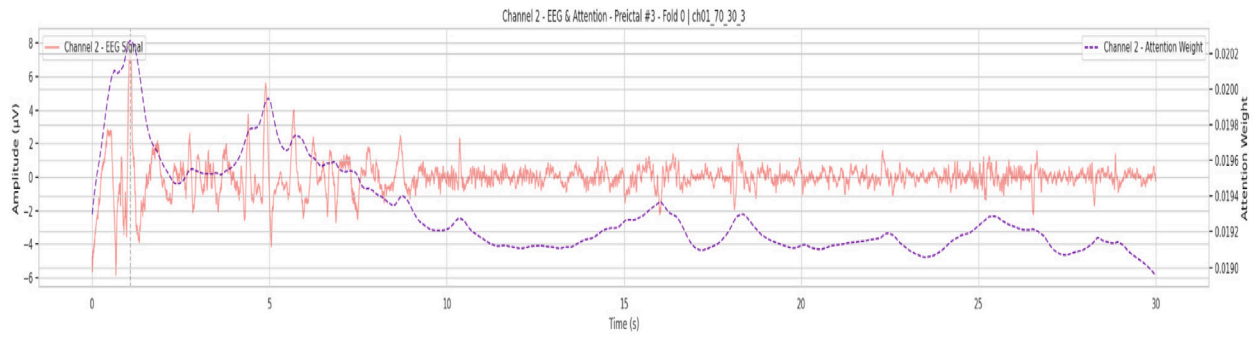


Fig. 12. Preictal EEG segment (Channel 2) with attention weights. The attention peaks align with sharp deflections and oscillatory bursts in the EEG, suggesting the model's focus on early preictal patterns.

Figs. 12 and 13 illustrate this mechanism. In preictal EEG segments, attention weights (purple dashed line) peak around sharp deflections and oscillatory bursts in the raw EEG (red line) patterns commonly linked to pre-seizure onset. In contrast, during interictal periods, attention remains low and aligned with stable background activity (orange dashed line), reinforcing the model's ability to neglect isolated spike-sharp and noise artifacts. Importantly, this attention-driven behavior not only boosts performance but also improves interpretability. In clinical contexts, the visualization of attention weights provides temporal markers that neurologists can cross-reference with visual EEG patterns. This bridges the gap between black-box models and clinical trust.

In summary, this section demonstrates how combining architectural innovations with a structured, automated MLOps process can substantially improve both model performance and clinical relevance in seizure prediction.

6.2.7. Evaluation of model generalization and cross-patient transferability

In addition to the high performance shown in CV strategies, our model is also able to attain good performance in holdout datasets that contain unseen data. Table 9 shows on average an accuracy of 0.987, a sensitivity of 0.994, and a specificity of 0.981, with a low FPR of 0.019. These results are a direct consequence of our good practices, which include 5-Fold CV from scratch, early stopping, and dynamic learning rate scheduling. Together with the CNN-BiLSTM-Attention architecture, the model avoids overfitting and promotes generalization to unseen EEG segments.

Our work is founded upon a patient-specific training paradigm. By being tailored to each subject's unique EEG pattern. Aside from patient-personalized optimization, we also examined the model's capacity to generalize to new, unseen patients. This consideration is important for clinical deployment, particularly in environments where patient history is unavailable. A model that can already perform well on the first EEG recording has clear practical advantages, such as immediate usability and reduced dependence on retraining. As a consequence, we employed two complementary validation schemes:

- 5-Fold Cross-Validation (CV): Training and validation were conducted on patients ch01, ch02, ch03, ch05, and ch06, with testing done on both a holdout subset and on the unseen patient ch08.
- Leave-One-Patient-Out (LOPO): Training was conducted on ch01–ch06, with ch08 entirely held out only for final testing.

Table 9 presents the results for both strategies. For the 5-Fold CV configuration, the model achieved an accuracy of 0.97, a sensitivity of 0.98, and FPR of 0.03 on the train and holdout sets. When tested on an unseen patient (ch08), the performance dropped but remained high with an accuracy of 0.9130 and a sensitivity of 0.9206. Similarly, under the LOPO protocol, the model still had high accuracy

Table 9

Cross-patient performance metrics under 5-Fold CV and LOPO strategies.

Metric	5-Fold CV			LOPO	
	Train/Val	Test holdout	Test on ch08	Train/Val	Test on ch08
Accuracy	0.9734	0.9775	0.9130	0.953	0.951
Sensitivity	0.9852	0.9856	0.9206	0.962	0.9629
Specificity	0.9617	0.9694	0.9054	0.944	0.9471
FPR	0.0383	0.0306	0.0426	0.038	0.0313

(0.9510) and sensitivity (0.9629) on unseen patient ch08, with an FPR of 0.0313, demonstrating strong generalization even under the most rigorous evaluation conditions.

These results confirm that while our model is trained specifically for a patient, the architecture of CNN, BiLSTM, and attention mechanisms in tandem learn and generalize EEG representations that transfer across subjects. This dual functionality makes this model ideal for both long-term, personalized monitoring and real-world clinical use where adaptability, flexibility, and reliability are key.

6.3. Evaluation of inference models in real production environment

This section presents the production performance results across three key dimensions: inference latency and throughput, system robustness, and monitoring stability. These metrics are essential for assessing real-time readiness, where low-latency response, high availability, and consistent behavior are critical for clinical deployment.

6.3.1. Inference time and throughput performance

Thanks to MLOps integration, each inference model was deployed as a separate containerized endpoint using TensorFlow Serving, enabling parallel request handling through buffering and queuing. Importantly, all tests were conducted remotely using REST API requests over the internet, not within a local network.

In the first test scenario, we monitored the inference performance of a single model instance dedicated to patient ch01. As shown in Fig. 14, the model maintained a stable serving state (UP) with an average inference time of 355 ms per request. The system achieved an average serving RPS or throughput, of approximately 1.1 RPS, matching the client's rate of 1.2 RPS. This latency is already sufficient for real-time clinical use, especially considering that a prediction requires the analysis of a full 30-second EEG segment.

In the second scenario, we stress-tested the infrastructure by launching inference across all model patients simultaneously. As illustrated in Fig. 15, the system scaled to handle a combined client-side request load of 21.8 requests per second, with the server sustaining a throughput of 16.3 RPS. All 19 model instances remained in a healthy state UP throughout the test. Under this increased load, the average inference

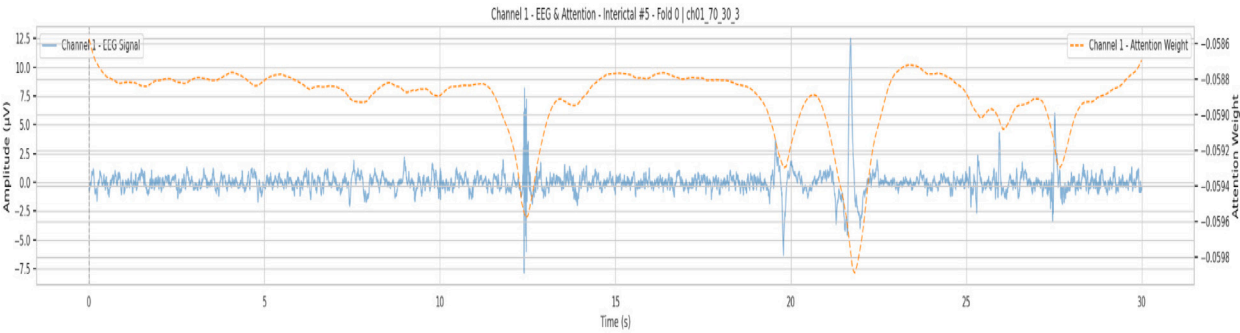


Fig. 13. Interictal EEG segment (Channel 1) with attention weights. The model gives higher attention to flat, stable background activity and lower attention to spikes or sharp transients, showing its ability to ignore non-informative patterns.

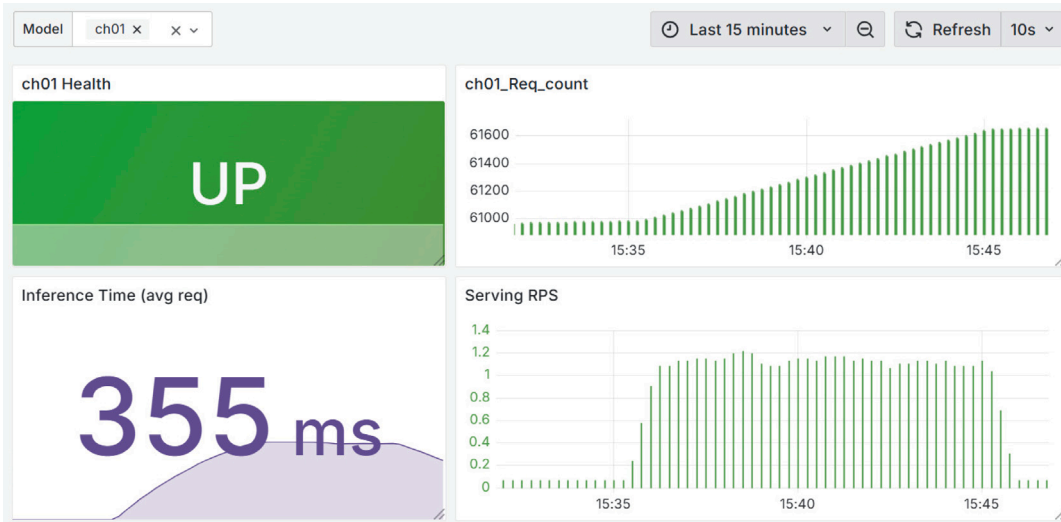


Fig. 14. Grafana monitoring of the ch01 model. The service is operational (UP) with an average inference latency of 355 ms, a 1.2 stable RPS, and a continuously increasing request count over time.

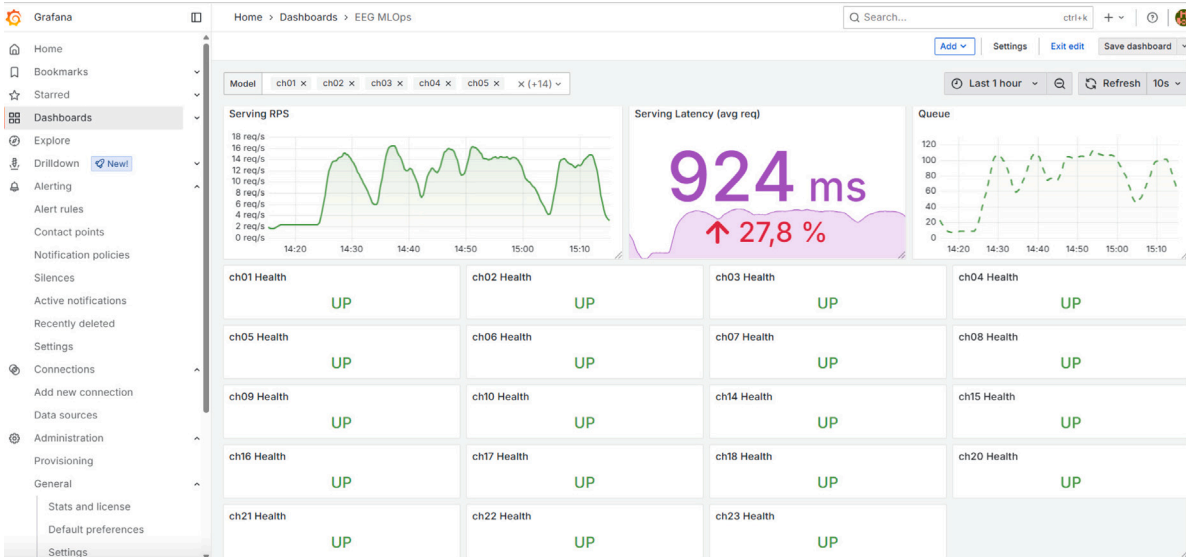


Fig. 15. Global Grafana dashboard displaying the status of models (ch01 to ch23). The system maintained full operational availability while sustaining a high request throughput under concurrent load with an inference time of less than 1 s.

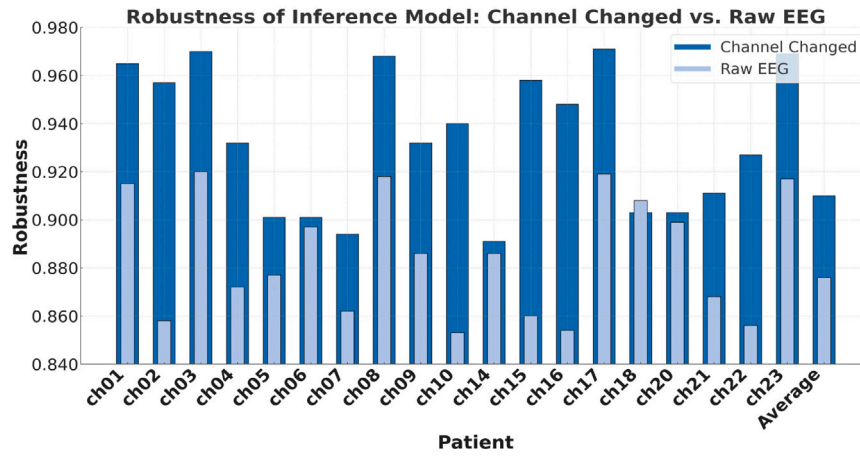


Fig. 16. Robustness comparison between inference using alternative EEG channels and raw unfiltered EEG data across all patients.



Fig. 17. First Panel for real-time epilepsy prediction (Real use-case ch02).

time rose to 924 ms. This latency is acceptable for real-time seizure prediction. However, a significant queue buildup was observed around 100 requests, which highlights high demand.

These results confirm the scalability and reliability of our MLOps infrastructure. Key enablers include Docker-based isolation, automatic scaling, request buffering, and reproducible deployment pipelines. The ability to deliver consistent inference performance, both in isolated and multi-model deployment scenarios, demonstrates that our system is well-suited for real-time clinical EEG applications and can adapt to operational pressures without compromising accuracy or responsiveness.

6.3.2. Analysis of inference model robustness under real-time conditions

Fig. 16 illustrates the robustness of our deployed inference model in two real-time conditions: when EEG signals come from different channels than originally selected, and when raw EEG data is used without any preprocessing. Despite these challenges, the model presents moderate drift. For most patients, accuracy remains around 0.90 and 0.84 for changed channels and raw EEG respectively. This confirms the model's stability when faced with real-world conditions. These results confirm again the power of MLOps in delivering robust, production-ready models.

6.3.3. Monitoring deployed models using Prometheus and Grafana

In our implementation, each inference endpoint is monitored in real time using Prometheus for metric collection and Grafana for dashboard visualization. In addition to inference key metrics, Grafana dashboards check the healthiness of models (Fig. 15). Furthermore, configured alerts detect anomalies like model health issues, increased latency, or performance drops, enabling early problem identification. This supports MLOps goals of proactive maintenance and reduces manual checks. Prometheus and Grafana enhance transparency and reliability in ML systems.

6.4. Real-time serving dashboard

MLOps also emphasize operational visibility for stakeholders. To that end, a real-time serving dashboard provides both inference output (predictions and alarms) and live evaluation if ground-truth labels are available. Our dashboard offers a clear view of prediction and evaluation. Fig. 17 illustrates the first panel for prediction and alarm. The prediction heatmap uses blue for normal EEG segments and orange for seizure warnings. The second heatmap shows alarm generation. An alarm is triggered if 8 out of 10 segments are preictal. Blue indicates no alarm, red signals an upcoming seizure. This enhances reliability and accuracy, reducing false alarms.

Fig. 18 shows the second panel for real-time monitoring of the seizure prediction model's accuracy and performance when true labels are available. It includes a confusion matrix comparing predicted vs. true labels for interictal and preictal states, along with prediction times and error rates.

Overall, these evaluation components demonstrate the effectiveness of our MLOps workflow not only in deploying and serving accurate models but also in maintaining visibility, reliability, and responsiveness throughout the production lifecycle.

7. Conclusion and perspectives

This paper presents a comprehensive MLOps workflow for EEG-based epilepsy prediction, automating the process from data ingestion to real-time model deployment and monitoring. Using the CRISP-ML(Q) methodology, we established a robust framework that integrates software engineering practices with the deep learning model lifecycle, bridging the gap between research and clinical applications. Our 1D CNN-BiLSTM model with attention is patient-specific. It is tuned to increase prediction accuracy and manage changes in the brain before a seizure. The framework can successfully generalize unknown data

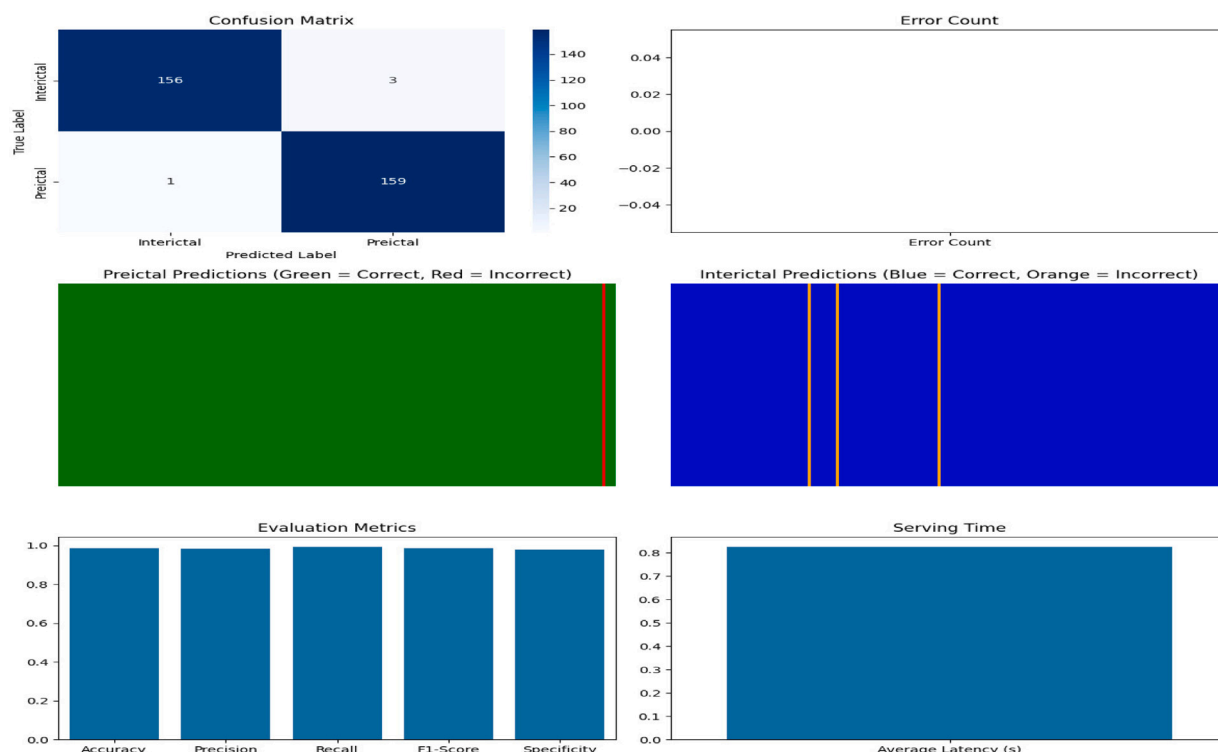


Fig. 18. Second Panel for evaluation real-time epilepsy prediction (Real use case ch02).

or patients when combined with thorough cross-validation and per-patient optimization. The pipeline automates preprocessing, training, and deployment to boost reliability, ensure consistency, and reduce errors. At the same time, using MLFlow and Grafana to integrate experiment tracking and real-time monitoring enhances transparency and maintains the system's healthiness. This MLOps best practice to automate pipelines marks a major step in improving patient live-care. It is a scalable and efficient AI-based solution for epilepsy management in the healthcare domain.

From a future-oriented perspective, the incorporation of explainable AI (XAI) techniques could help close the gap between neurologist interpretations and deep learning decisions during the modeling process. Eventually, local or global XAI modules can provide physicians with patient-by-patient explainable information about the black-box model's logic, which can increase clinical adoption and trust. In parallel, future work should focus on reducing model complexity to support deployment on low-power edge devices. This can be done by using lightweight architectures or techniques like pruning, quantization, and knowledge distillation. These improvements would allow real-time seizure prediction directly on the device, with faster response, better privacy, and continuous monitoring for remote or mobile healthcare.

CRediT authorship contribution statement

Salem Trabelsi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Salah Gontara:** Writing – review & editing, Software. **Khaled Ben Khalifa:** Validation. **Abdelatif Mtibaa:** Supervision.

Ethical approval

There is no any ethical conflicts.

Declaration of competing interest

The authors declare no conflict of interest.

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Data availability

The CHB-MIT Scalp EEG Database is available on the official page: <https://physionet.org/content/chbmit/1.0.0/>.

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